Atomic Structure

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Atomic Structure

LESSON-1

- DALTON'S ATOMIC THEORY
- DISCOVERY OF ELECTRON
- DISCOVERY OF PROTONS – POSITIVE RAYS (CANAL RAYS)
- SPECIFIC CHARGE

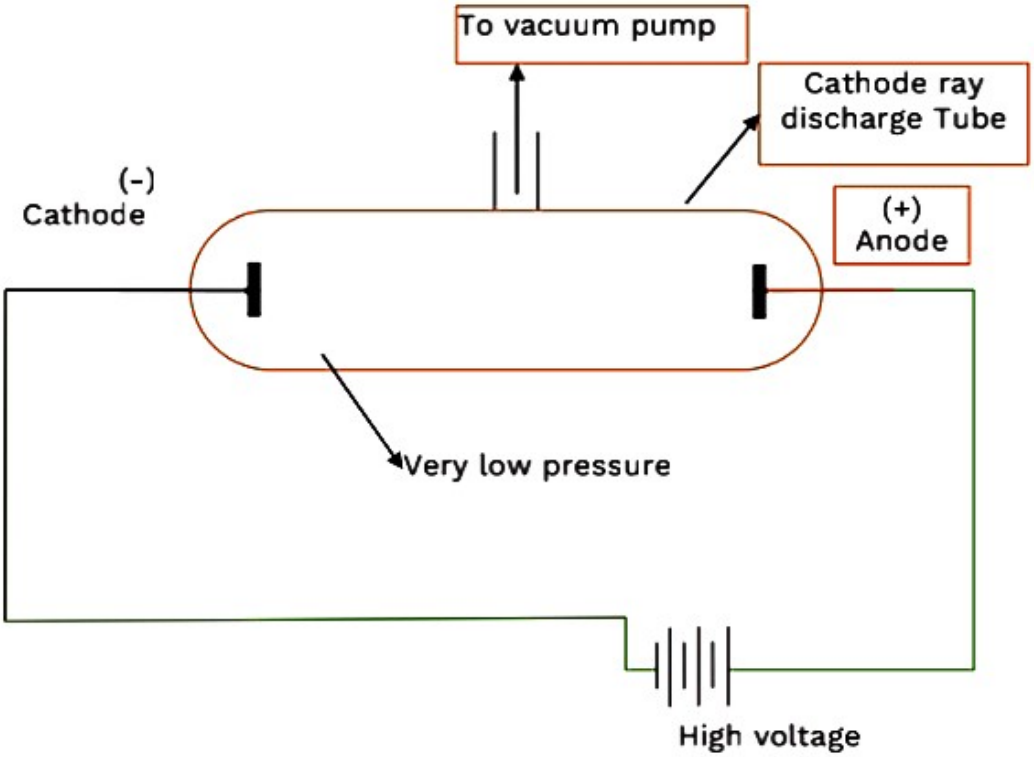
DALTON'S ATOMIC THEORY

The ultimate particle (smallest) of any matter is atom (indivisible).

After discovery of subatomic particles, this theory failed.

Discovery of electron

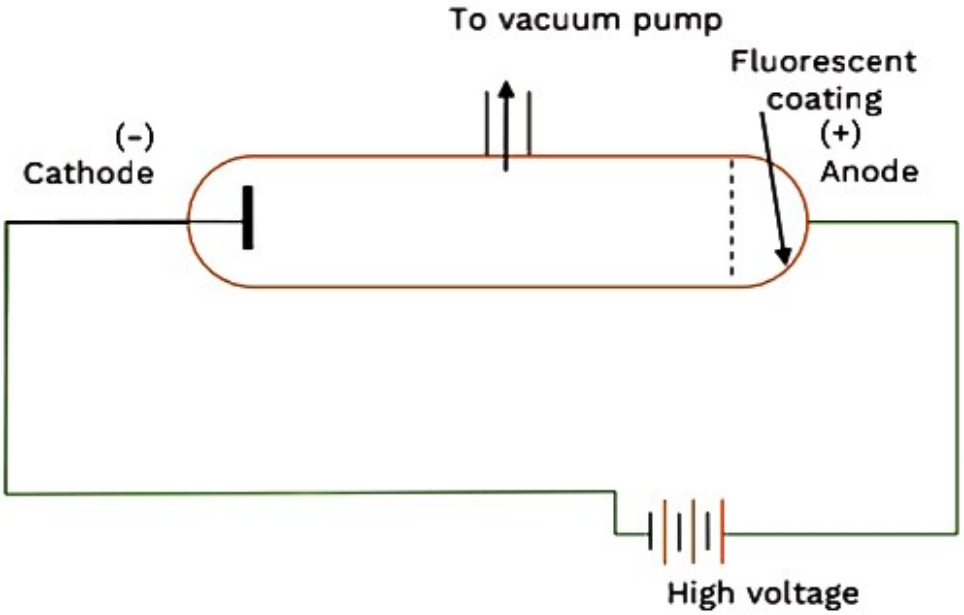
Based on Michael Faraday's cathode ray experiment, it was concluded that electrons are basic constituent of all the atoms.



This experiment was done in 1830. In this experiment, when a sufficient high voltage is supplied, then current starts flowing through a stream of particles moving in the tube from cathode to anode. These stream of particles are called **cathode rays**.

Important observations

1. The cathode rays start from cathode and move towards the anode.
2. These rays themselves are not visible but their behaviour can be observed with the help of certain kind of materials (Fluorescent or phosphorescent) which glow when hit by them.



3. In the absence of electrical or magnetic field, these rays travel in straight lines.
4. In the presence of electrical or magnetic field, the behaviour of cathode rays are similar to that expected from negatively charged particles, suggesting that the cathode rays consist of negatively charged particle, called electrons.
5. Produce mechanical effect, as they can rotate the wheel placed in their path.
6. The characteristics of cathode rays (electrons) do not depend upon the material of electrodes and the nature of gas present in the cathode ray tube.

DISCOVERY OF PROTONS – POSITIVE RAYS (CANAL RAYS)

a) Goldstein (1886) repeated the discharge tube experiment but he used perforated cathode and noticed the emission of positive rays or canal rays.

Note: These rays do not originate from anode, and so it is wrong to call them anode rays.

b) The specific charge (e/m) of canal rays particles varied with nature of gas and was found to be maximum if H_2 was used.
c) The positive rays particles were thus, called positively charged gaseous atoms left after the removal of electron or ionized gaseous atoms. However, if H_2 gas is used in discharges, the positive rays particles are named as protons (usually represented as P).
d) Thus, a subatomic particle, that is a fundamental constituent of all matter, is called proton; it has mass 1.672×10^{-27} kg and charge + 1.603×10^{-19}C.
e) Behaviour in electric and magnetic field is just opposite to cathode rays.

	Mass of one particle	Type of charge	Charge on one particle
Proton (P)	1.672×10^{-27} kg	Positive	$+e(1.6\times10^{-19}C)$
Neutron (n)	1.674×10^{-27} kg	Neutral	Zero
Electron(e)	9.1×10^{-31} kg	Negative	$-e$
Alpha particle(α)	$2\ m_p+2m_n$	Positive	$+2e$

SPECIFIC CHARGE

For a particle,

$$\text{Specific charge} = \frac{\text{Charge on one particle}}{\text{Mass of one particle}} \qquad \text{Unit}: \frac{C}{kg}$$

Q1 **Determine specific charge of electron.**

A1 $$\text{Specific charge} = \frac{1.6 \times 10^{-19} C}{9.1 \times 10^{-31} kg} \simeq 1.7 \times 10^{11}\ Ckg^{-1}$$

Q2 **Ratio of specific charge of proton with an alpha - particle is:**
(A) 1:2 (B) 2:1 (C) 1:4 (D) 4:1

A2 $$\frac{\text{Specific charge of proton}}{\text{Specific charge of } \alpha} = \frac{q_1 / m_1}{q_2 / m_2} = \frac{e / m_p}{2e / (2m_p + 2m_n)} = \frac{1}{2} \times \frac{4m_p}{m_p} = \frac{2}{1}$$

$\because m_p \simeq m_n$

$\therefore m_\alpha = 4m_p$ or $4m_n$

Q3 **Which of the following particle has highest value of specific charge?**
(A) α-particle (B) Proton (C) Neutron (D) Electron

A3 **D**

α	proton	neutron	electron
$\frac{2e}{4m_p}$	$\frac{e}{m_p}$	0	$\frac{e}{m_e}$

$$\frac{1}{2m_p} : \frac{1}{m_p} : \frac{1}{m_e}$$

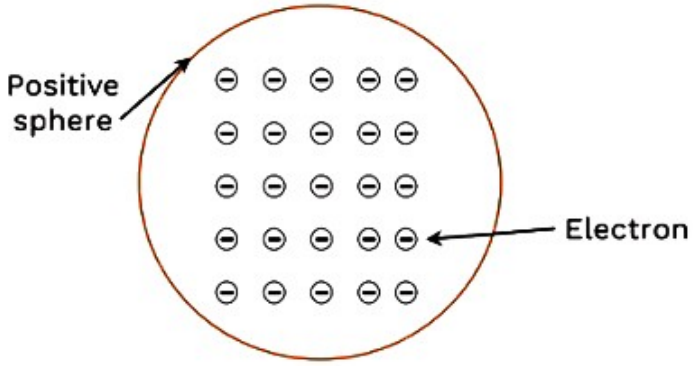
Thomson Model

LESSON-2

- THOMSON MODEL
- RUTHERFORD'S α-PARTICLE SCATTERING EXPERIMENT
- RUTHERFORD'S ATOMIC MODEL

FIRST ATOMIC MODEL

An atom is spherical in which total mass and positive charge is uniformly distributed throughout the sphere and electrons are embedded into it in such a manner so as to give it maximum stability.

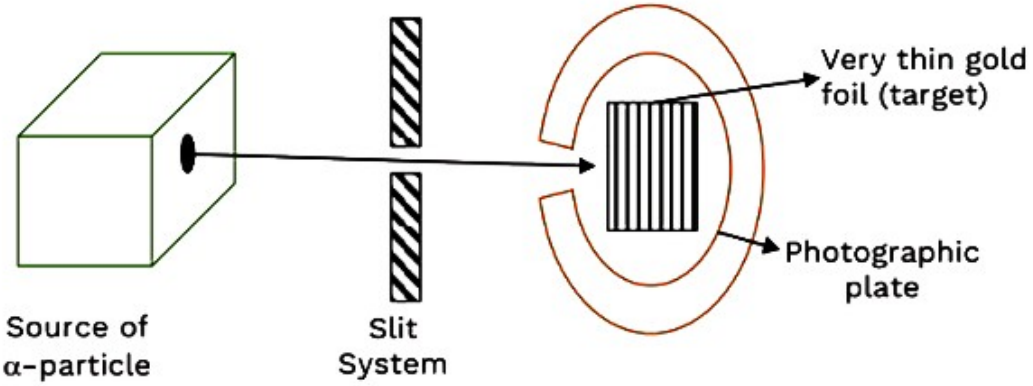


⇒ This model is also known as **Plum Pudding model or water melon model.**

⇒ This model was rejected after results of Rutherford's experiment.

Rutherford's α–particle scattering experiment

Set up:



Fast moving α particle are emitted from the source which are directed towards gold foil.

After striking on the foil, the path of α–particles can be traced by flashes produced on the photographic plate.

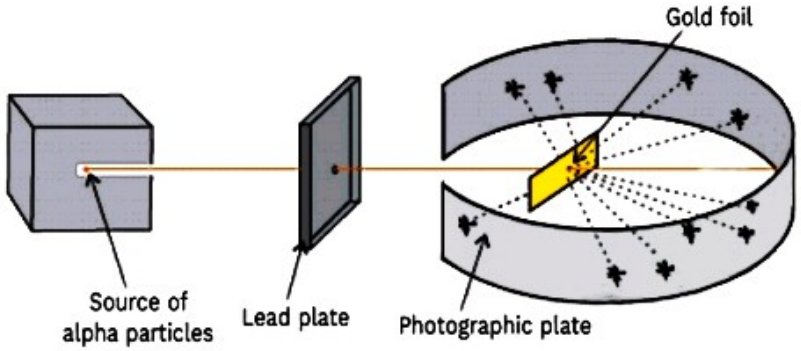
OBSERVATIONS:

i) Most α–particle are passed straight through the foil without any deflection.

ii) Few α–particles are deflected by angles like 60°, 90°, 110° etc.

iii) Very few (almost one in very 20,000) particles are deflected by almost 180°.

iv) Angle of deflection increases with increase in atomic number of target metal.



Conclusion

i) Most part of the atom is empty.

ii) Total positive charge and most of the mass is concentrated in a very small region called as **Nucleus**.

iii) The size of nucleus is very–very small as compared to atom.

RUTHERFORD'S ATOMIC MODEL

i) Nucleus is present at the centre of atom.

ii) Electrons are present outside the nucleus and they revolve around the nucleus with very high speeds.

iii) The atom is stable due to electronic attraction between electrons and nucleus.

iv) Radius of atom $\simeq 10^5$ radius of nucleus.

v) For an elements,

$$\boxed{r_{nucleus} = r_0 A^{1/3}}$$

Where $r_0 = \frac{4}{3} \times 10^{-13}$ cm (constant)

A = mass number of element

Rutherford's Model Continued

LESSON–3

- RUTHERFORD'S MODEL CONTINUED
- DISTANCE OF CLOSEST APPROACH (r_{min})
- PROBLEMS ON RUTHERFORD'S MODEL
- DRAWBACK OF RUTHERFORD'S MODEL

Mathematical part

1. **For any particle,**

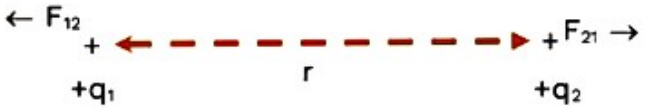
$$\boxed{\text{Total energy} = \text{K.E} + \text{P.E}}$$

Where K.E. → Kinetic energy of particle (due to motion) i.e. $\frac{1}{2}mv^2$

P.E. → Potential energy of particle (due to position in a conservative force)

2. **Coulomb's law**

This law is used to determine electrostatic force between two charged particles.



$$|F_{12}| = |F_{21}|$$

$$\boxed{\text{Electrostatic force} = k\frac{q_1q_2}{r^2}}$$

Where k = constant (depends upon medium) = $9 \times 10^9 \frac{Nm^2}{C^2}$

q_1 = charge on first particle
q_2 = charge on second particle
} in Coulomb

r = distance between both particles (in meter)

3. **Potential energy due to electrostatic force**

P.E = Force × Distance

$$\boxed{P.E = \frac{Kq_1q_2}{r}}$$

For like charges: P.E > O

If value of r increases then P.E decreases.
If value of r decreases then P.E increases.

For unlike charges: P.E < O

If r increases, then P.E. increases.
If r decreases, then P.E decreases.

Distance of closest approach (r_{min})

When the α-particle reaches to the closest distance to the nucleus, it will come to rest and its intial kinetic energy will be completely transformed into potential energy.

r_{min} is determined for that α–particle which gets deflected by 180°.

When an α–particle having K.E $= \frac{1}{2}mv^2$ enters into the electric field of the nucleus, its kinetic energy starts converting into its P.E.

As distance decreases, the P.E increases so kinetic energy will decrease

At a certain point, total K.E is converted into P.E. It means at this point, the α–particle comes at rest.

This is the closest distance from the nucleus which is approached by α–particle, hence it is called r_{min}.

So at $r=r_{min}$,

Initial $(K.E)_\alpha = (P.E)_\alpha$ at closest approach.

$$\frac{1}{2}mv^2 = \frac{K(2e)(Ze)}{r_{min}}$$

$$r_{min} = \frac{4KZe^2}{mv^2}$$

Where m = Mass of α–particle (kg)

v = Speed of α–particle (ms^{-1})

K.E. = Initial K.E of α–particle (Joule)

$K = 9\times10^9\ N.m^2/C^2$

$e = 1.6\times10^{-19}\ C$

Z = Atomic no. of target metal

r_{min} = Distance of closest approach (m)

Note : eV = electron volt (unit of energy)

$1eV = 1.6 \times 10^{-19}$ Joule

Q4 **An α–particle travels with 8.64 MeV energy towards the nucleus of a metal atom (Z=30) and arrives at a distance of 10^{-x} nm from the nucleus, then value of 'x' is:**
[Given: e=1.6×10^{-19} C, K= 9×10^{9} N.m^2/C^2]
(A) 14 (B) 10 (C) 5 (D) 11

A4 **C**

Given, K.E of α–particle = 8.64×10^{6} eV

$= 8.64\times10^{6} \times 1.6\times10^{-19}$ Joule

$r_{min} = 10^{-x}$nm

$= 10^{-x} \times 10^{-9}$m

$= 10^{-(x+9)}$m

Since at r_{min},

$$K.E = P.E = \frac{K(2e)(Ze)}{r_{min}}$$

$$r_{min} = \frac{K(2e)Ze}{K.E}$$

$$10^{-(x+9)} = \frac{9\times10^{9}\times2\times1.6\times10^{-19}\times30\times1.6\times10^{-19}}{8.64\times10^{6}\times1.6\times10^{-19}}$$

$$10^{-(x+9)} = \frac{864\times10^{9-19-19-6+19}}{8.64}$$

$$= 10^{2} \times 10^{9-25}$$

$$= 10^{2-16}$$

$$= 10^{-14}$$

Bases are same, powers are equal

$$-(x+9) = -14$$

$$x + 9 = 14$$

$$x = 14 - 9$$

$$x = 5$$

Q5 **In Rutherford experiment, if α–particle is projected with v_0 velocity towards nucleus of an atom of atomic number 'Z' its closest distance of approach is R, then what is the closest distance of approach, if α–particle with $4v_0$ velocity is projected towards nucleus of atom of atomic number "2Z"**

(A) R **(B) R/16** **(C) R/8** **(D) R/2**

A5 **(C)**

$$K.E = \frac{K(2e)(Ze)}{r_{min}}$$

$$\text{or } \frac{1}{2}mv^2 = \frac{K(2e)(Ze)}{r} \qquad \left[\text{here } \frac{1}{2}m, K, \text{ and } e \text{ are constants}\right]$$

$$r \propto \frac{Z}{v^2}$$

$$\frac{r_1}{r_2} = \frac{Z_1}{Z_2}\left(\frac{v_2}{v_1}\right)^2$$

According to Question

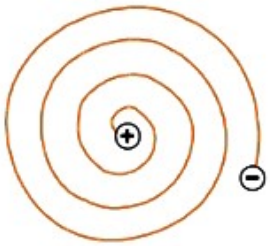
$$\frac{R}{r_2} = \frac{Z}{2Z}\left(\frac{4v_0}{v_0}\right)^2$$

$$r_2 = \frac{R}{8}$$

Drawback of Rutherford's model

i) According to classical electrodynamics theory given by Maxwell, it suggests that if a charged particle moves under acceleration then it emits energy continuously.

ii) According to Rutherford's Atomic Model, electron revolve around the nucleus and hence have acceleration due to circular motion. So, the electron moves under acceleration so it should emit energy and collapse into the nucleus following spiral path.



But this does not happen.

iii) It is unable to explain electronic structure of atom (No idea about position and energy of e^-).

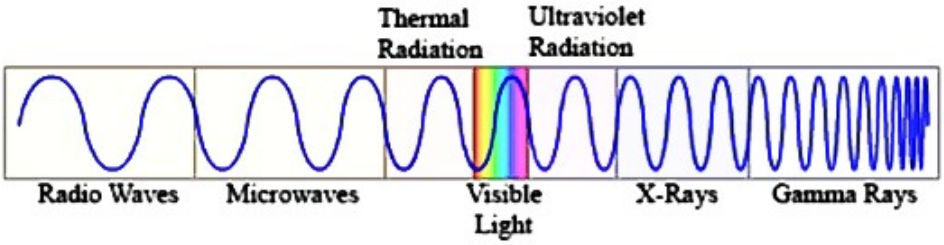
iv) It could not explain discontinuous spectrum.

WAVE

A wave is a mode of transferring energy from one point to another point without actual transfer of matter.
e.g. water waves, shock waves, electromagnetic radiation.

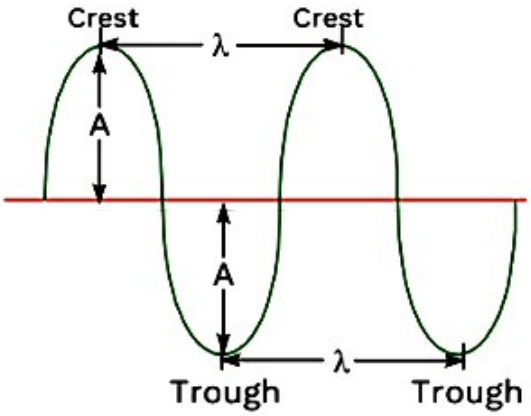
LESSON-4

- MAXWELL'S THEORY OF ELECTROMAGNETIC RADIATION (EMR)
- EMR SPECTRUM
- WAVE



Characteristics of waves

Amplitude (A) is the height of crest or depth of trough.



Speed of wave (c)
A wave always moves with constant speed.
Wavelength (λ):
Length of one complete wave.
In other words,
Wavelength is the distance between two successive crests (or troughs).

SI unit is meters

Other units are cm, mm, Å , nm, pm, etc

$$\boxed{\begin{aligned} 1\,\text{Å} &= 10^{-10}\,\text{m} \\ 1\,\text{nm} &= 10^{-9}\,\text{m} \\ 1\,\text{pm} &= 10^{-12}\,\text{m etc} \end{aligned}}$$

Frequency (ν)

The number of oscillations (waves) in one second from a fixed point.

SI unit is per second (sec^{-1}) or Hertz (Hz)

Note: Let time = 1 sec

No. Of waves = ν

Distance travelled = $\nu \times \lambda$

We know that:

$$\text{speed} = \frac{\text{distance}}{\text{time}}$$

$$c = \frac{\nu \times \lambda}{1}$$

$$\boxed{c = \nu \times \lambda}$$

Wave Number $(\bar{\nu})$

Number of waves in one metre length.

$$\boxed{\bar{\nu} = \frac{1}{\lambda}}$$

SI unit of wave number is m^{-1}

Time Period (T)

The time taken by the wave in one complete oscillation.

$$\text{Time taken} = \frac{\text{distance}}{\text{speed}}$$

$$= \frac{\lambda}{c}$$

$$= \frac{\lambda}{\nu\lambda} \qquad [\because c = \nu\lambda]$$

$$\boxed{T = \frac{1}{\nu}} \text{ second}$$

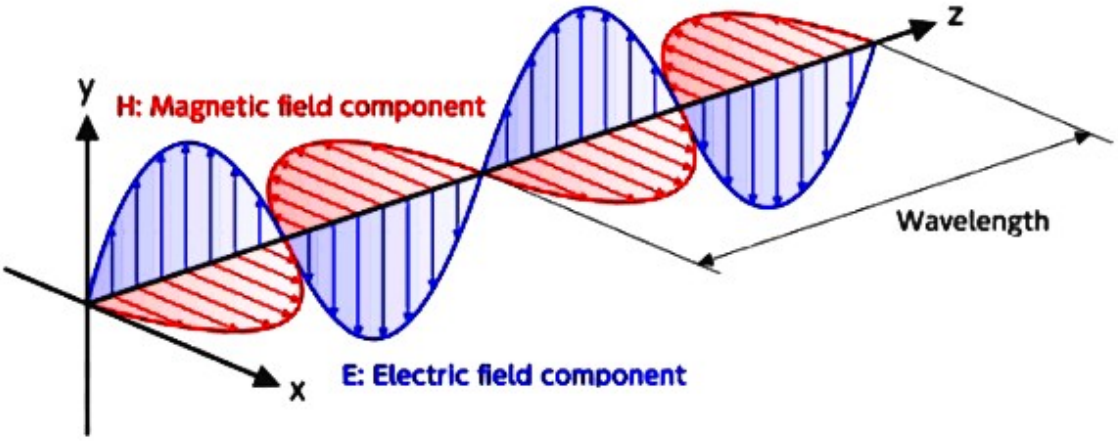
Maxwell's Theory of Electromagnetic Radiation (EMR)

i) It suggests wave nature of EMR.

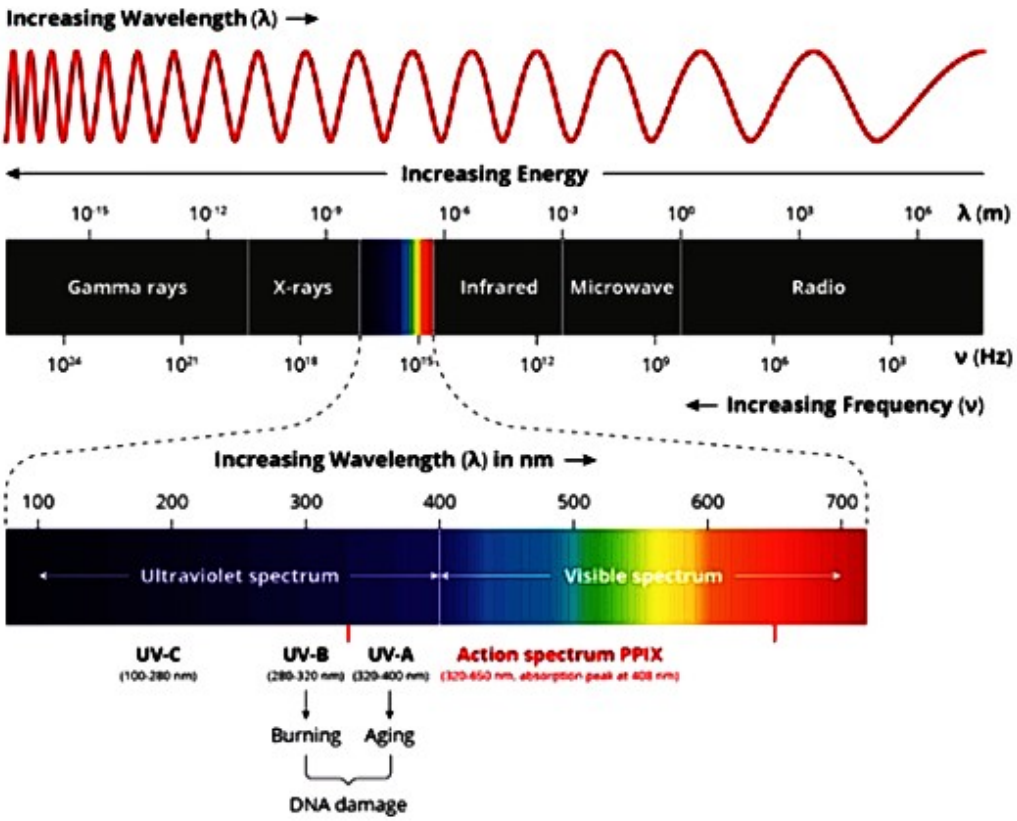
ii) When a charged particle moves under acceleration, alternating electric and magnetic fields are produced.

iii) These fields are in phase and oscillating mutually perpendicular to each other.

iv) The direction of energy propagation is perpendicular to both fields.



EMR Spectrum



v) All EMR travel with speed of light. ($c = 3\times10^{8}\ ms^{-1}$)

$\therefore$ speed of wave = $\nu \times \lambda$

So, for EMR

$$c = \nu \times \lambda \Rightarrow \nu = \frac{c}{\lambda}$$

vi) In EMR, both electric and magnetic components have same λ, ν, $\bar{\nu}$,T,A etc.

Note:

- An EMR does not need a medium to travel means they can travel in vacuum also.
- All EMR show diffraction and interference which are characteristic properties of waves so this also proves that all EMR travel as waves.

Q6 **Radiocity broadcasts on a frequency of 5,090 kHz. What is the wavelength of electromagnetic radiation emitted by the transmitter?**

(A) 10.3 m **(B) 58.9m** **(C) 60.5 m** **(D) 75.5 m**

A6 **(B)**

Given : Frequency $= 5090\ \text{kHz}$

$= 5090 \times 10^3\ \text{Hz}$

We known that :

$$c = \nu\lambda$$

$$\lambda = \frac{3 \times 10^8\ \text{ms}^{-1}}{5090 \times 10^3\ \text{Hz}}$$

$$\lambda = \frac{3 \times 10^{8-3}}{5090}$$

$$\lambda = \frac{3 \times 10^5}{5090}$$

$$\lambda = 58.9\ \text{m}$$

Q7 **A source of electromagnetic radiation emits monochromatic radiation of 50 nm, then what is the wavenumber $(\bar{\nu})$ of this radiation?**

(A) $2\times10^8\ m^{-1}$ **(B) $2\times10^7\ m^{-1}$** **(C) $5\times10^{-8}\ m^{-1}$** **(D) $5\times10^{-9}\ m^{-1}$**

A7 **(B)**

Given : Wavelenghth, $\lambda = 50\ \text{nm}$

$= 50 \times 10^{-9}\ \text{m}$

Wavenumber, $\bar{\nu} = \frac{1}{\lambda}$

$$= \frac{1}{50 \times 10^{-9}}\ \text{m}^{-1}$$

$$= 2 \times 10^7\ \text{m}^{-1}$$

Planck's Quantum Theory

LESSON–5

- PLANCK'S QUANTUM THEORY
- PROBLEMS ON PLANCK'S QUANTUM THEORY

i) It suggests particle like nature of EMR.

ii) Planck suggested that atoms or molecules absorb or emit energy in discrete manner not in continuous manner.

iii) It means the exchange of energy by a substance as EMR takes place in form of packets / bundles.

iv) These packets of energy are known as quantum or quanta.

v) All quantum of a particular EMR are identical means contain same amount of energy.

vi) Quantum are massless (Just packet of energy).

vii) Quantum of light is also known as photon.

For a particular EMR

Energy of quantum (E) $\propto$ frequency of radiation (ν)

$$\boxed{E = h\nu}$$

or

$$\boxed{E = \frac{hc}{\lambda}} \qquad \boxed{\because c = \nu\lambda \Rightarrow \nu = \frac{c}{\lambda}}$$

Where h is Planck's constant (6.6×10^{-34} Js)
ν is frequency (s^{-1})
c is speed of light (3×10^{8} ms^{-1})
λ is wavelength of EMR (m)
E is energy of quantum (J)

viii) Since quantum is the smallest packet of energy so the energy absorbed or released by a body or substance as EMR will be always an integral multiple of $h\nu$
i.e. $h\nu$, $2h\nu$, $3h\nu$, ...

So total energy absorbed / released by substance as EMR = $Nh\nu$ or $Nh\frac{c}{\lambda}$

Where N = no of quantums.

ix) The energy of one single quantum is very small which is generally measured in eV(electron volt)

$$\boxed{1\ eV = 1.6\times10^{-19}\ J}$$

Also $$\boxed{(E)_{Quantum} \simeq \frac{1240}{\lambda(nm)}\ eV}$$

ILLUSTRATION:

Q8 **The energy content per quantum (Joule) for a radiation of frequency 4.2×10^{14} Hz will be**

(A) 2.8×10^{-21} **(B)** 2.5×10^{-19} **(C)** 2.8×10^{-19} **(D)** 2.5×10^{-18}

A8 **(C)**

Given: frequency $= 4.2\times10^{14}$ Hz

$h = 6.6\times10^{-34}$ Js

We know $E = h\nu$

$= 6.6\times10^{-34} \times 4.2\times10^{14}$

$= 27.72\times10^{-20}$

$= 28\times10^{-20}$ J

$= 2.8\times10^{-19}$ J

Q9 **Energies of quanta of two different radiations are 2 eV and 4 eV respectively. Then their wavelength are related as:**

(A) $\lambda_1=\lambda_2$ **(B)** $2\lambda_1=\lambda_2$ **(C)** $\lambda_1=2\lambda_2$ **(D)** $\lambda_1=3\lambda_2$

A9 **(C)**

Let wavelength of 1st radiation $= \lambda_1$

Wavelength of 2nd radiation $= \lambda_2$

We know that : $E = h\nu$

$$E = \frac{hc}{\lambda}$$

$$\frac{E_1}{E_2} = \frac{h\frac{c}{\lambda_1}}{h\frac{c}{\lambda_2}} = \frac{\lambda_2}{\lambda_1}$$

$$\boxed{\lambda_1 = 2\lambda_2}$$

Q10 **In Astronomical observations, signals observed from the distant stars are generally weak. If the photon detector receives 3×10^{-18} J energy from radiations of 600 nm wavelength then calculate the number of photons received by the detector**
($h=6\times10^{-34}$ J.S)
(A) 10 **(B) 20** **(C) 30** **(D) 40**

A10 **(A)**

Given wavelength = 600 nm and Energy $= 3\times10^{-18}$ J

$$\text{Energy} = \frac{Nhc}{\lambda}$$

$$3\times10^{-18}\,J = \frac{N\times 6\times10^{-34}\times3\times10^{8}}{600\times10^{-9}}$$

$$N = 10 \text{ photon}$$

Q11 **The energy required to break one mole of Cl–Cl bonds in Cl_2 is 242 $kJmol^{-1}$. The longest wavelength of light capable of breaking a single Cl–Cl bond will be:**
(A) 594 nm **(B) 640 nm** **(C) 700 nm** **(D) 491 nm**

A11 **(D)**

1 mole Cl_2 = 1 mole Cl – Cl bond
$\cong 6\times10^{23}$ Cl – Cl bonds

$$\text{Energy for one Cl – Cl bond} = \frac{242\times10^{3}}{6\times10^{23}}\times1\,J$$

$$\simeq 40.3\times10^{-20}\,J$$

$$E_{Photon} \geq 40.3\times10^{-20}\,J$$

$$\frac{hc}{\lambda} \geq 40.3\times10^{-20}$$

$$\lambda \leq \frac{6.6\times10^{-34}\times3\times10^{8}}{40.3\times10^{-20}}\,m$$

$$\lambda = 491 \text{ nm}$$

Q12 **A 100 watt bulb emits monochromatic light of wavelength 300 nm, then number of photon emitted per second by the bulb is:**

(A) 1.5×10^{21} **(B) 1.5×10^{22}** **(C) 15×10^{21}** **(D) 1.5×10^{20}**

A12 **(D)**

Given : wavelength = 300 nm

We know that : $\text{Power} = \frac{\text{Energy(J)}}{\text{Time(s)}}$

Let time = 1 sec

then Energy = 100 J

$100\,J = \frac{Nhc}{\lambda}$ $\left(\begin{matrix}N = \text{no. of photons} \\ \text{emitted in 1 second}\end{matrix}\right)$

$N = \frac{100 \times 300 \times 10^{-9}}{19.8 \times 10^{-26}}$

$N = \frac{3 \times 10^{21}}{19.8}$

$N = 0.15 \times 10^{21}$

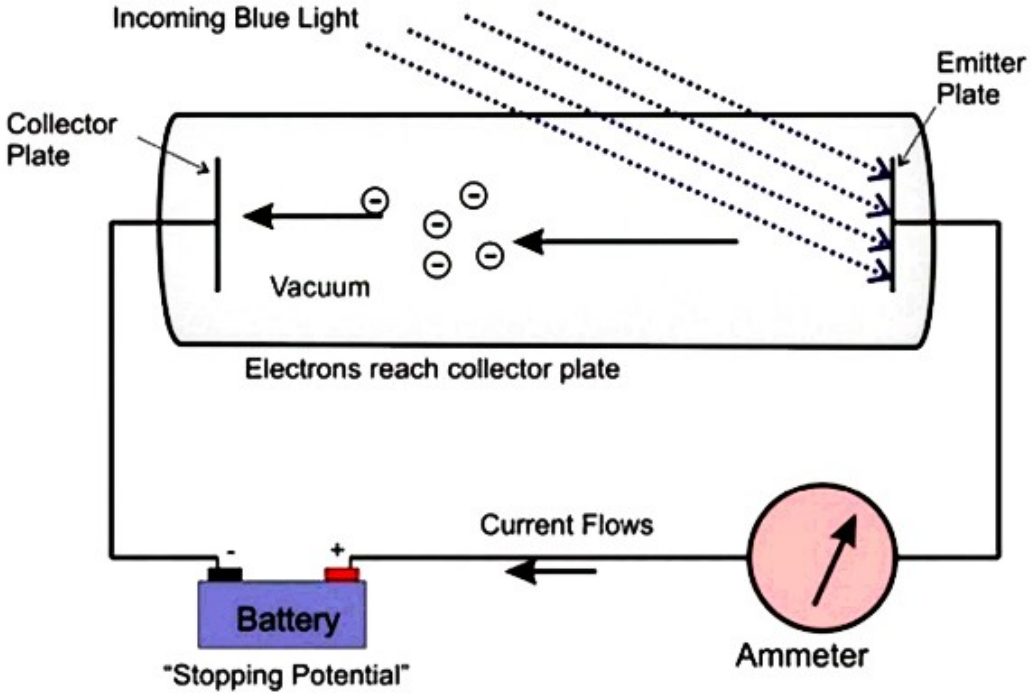
$N = 1.5 \times 10^{20}$

Photoelectric Effect

Photoelectric effect was performed by **H.Hertz.** When the metal surface is exposed to a beam of light of particular frequency (or wavelength), electrons are ejected from metal surface. This phenomenon is called **photoelectric effect**. Ejected electrons are called **photoelectrons** and the current flow due to this is called **photocurrent**.

LESSON–6

- PHOTOELECTRIC EFFECT
- PROBLEMS OF PHOTOELECTRIC EFFECT



OBSERVATIONS

i) There is no time lag between striking of light and ejection of electron from metal surface.

ii) As intensity of light increases, magnitude of photocurrent increases.

iii) For each metal, there is a certain minimum value of frequency of incident light is required below which photoelectric is not observed.

iv) As frequency of incident light increases, kinetic energy of photoelectron in the tube increases.

Explanations

i) Einstein explained this phenomenon using Planck's Quantum Theory.

ii) Electrons present in the metal need a certain minimum amount of energy to get free from the electrostatic attraction of the nucleus.

iii) The beam of light can be considered as the stream of energy carrying massless particles called photons.

iv) When a photon having sufficient amount of energy strikes with an electron in the metal, it transfers its energy instantaneously to the electron and the electron is ejected out from the metal surface simultaneously.

v) For ejection of one electron, only one photon is used.
So, energy of photon of incident light = energy utilized by electron to get free from metal + $(K.E)_{max}$ of ejected electron.

$$\boxed{E_{photon} = \phi + K.E_{max}}$$

(We take maximum because electrons present in outermost shell)

Where ϕ is work function of metal

If $E_{ph} < \phi \Rightarrow$ No ejection of e^- takes place

$E_{ph} = \phi \Rightarrow$ Ejection of e^- but $(KE)_{e^-} = 0$

$E_{ph} > \phi \Rightarrow$ Ejection of e^- and $(K.E)_{e^-} \neq 0$

Threshold frequency (ν_0)

The minimum frequency of incident photon to cause photo electric effect.
For photoelectric effect

$$\left(E_{photon}\right)_{min} = \phi$$

$$h\nu_0 = \phi$$

$$\Rightarrow \boxed{\nu_0 = \frac{\phi}{h}}$$ Constant for a particular metal

Now if 'ν' is the frequency of incident photon on a metal having threshold frequency (ν_0)

Cases:

i) If $\nu < \nu_0 \Rightarrow$ no PEE (photoelectric effect)

ii) If $\nu = \nu_0 \Rightarrow$ PEE but K.E = 0

iii) If $\nu > \nu_0 \Rightarrow$ PEE and KE $\neq$ 0

Also

$$KE = E_{ph} - \phi$$
$$= h\nu - \phi$$

i.e. If ν increases, then K.E of photoelectron increases.

Threshold wavelength

Wavelength corresponding to threshold frequency ν_0.

$$\boxed{\phi = h\nu_0 = \frac{hc}{\lambda_0}}$$

Q13 **The work function (ϕ) of some metals is listed below. The number of metals which will show photoelectric effect when the light of 300 nm wavelength falls on the metal.**

Metal	Zn	Na	K	Mg	Cu	Ag	Fe	Pt	W
ϕ eV	2.4	2.3	2.2	3.7	4.8	4.3	4.7	6.3	4.75

A13 $E_{Ph} = \frac{1240}{\lambda} = \frac{1240}{300} = 4.1eV$

For photo electric effect

$E_{Ph} \geq \phi$

$\Rightarrow \phi = 4.1eV$

Zn, Na, K, Mg will show photoelectric effect (4 metals)

Q14 **Electromagnetic radiations having 310 Å are subjected to a metal sheet having work function of 12.8 eV. What will be the velocity of photoelectrons with maximum kinetic energy :**

(A) 0, no emission will occur **(B) 2.18×10^6 m/s**

(C) $2.18\sqrt{2}\times10^6$ m / s **(D) 8.72×10^6 m/s**

A14 **(C)**

Given $\lambda = 310\,Å$ and $\phi = 12.8$ eV

$= 31$ nm

$$E_{Photon} = \frac{1240}{\lambda} = 40\,eV$$

So, $E_{Ph} > \phi$ Photoelectric effect

$KE_{max} = E_{Ph} - \phi$

$\frac{1}{2}mv^2 = 40eV - 12.8eV$

$\frac{1}{2}(9.1\times10^{-31})\times v^2 = 27.2eV$

$$v^2 = \frac{27.2\times1.6\times10^{-19}\times2}{9.1\times10^{-31}}J = \frac{43.52\times2}{9.1}\times10^{12}J$$

$= 9.564\times10^{12}$

$= 3.08\times10^6$ m / s $= 2.18\sqrt{2}\times10^6$ m / s

Photoelectric Effect Contd...

CONTENT

- PHOTOELECTRIC EFFECT CONTINUED
- INTENSITY OF RADIATION
- STOPPING POTENTIAL (V_0)

Q15 **What is the work function of the metal if the light of wavelength 4000 Å generates photoelectrons of velocity 6×10^5 ms^{-1} from it?**
(A) 3.1 eV (B) 2.1 eV (C) 4.0 eV (D) 0.9 eV

A15 **(B)**

Given, wavelength of light $= 4000\,\text{Å} = 400\,\text{nm}$

Velocity of photoelectron $= 6\times10^5\,\text{ms}^{-1}$

$$\text{K.E.} = \frac{1}{2}mv^2$$

$$= \frac{1}{2}\times\frac{9.1\times10^{-31}\times(6\times10^5)^2}{\left(1.6\times10^{-19}\right)}\,\text{eV} \ldots.[1\ \text{eV} = 1.6\times10^{-19}\,\text{J}]$$

$$= \frac{1}{2}\times\frac{9.1\times6\times6\times10^{10}\times10^{-31}}{1.6\times10^{-19}}$$

$$= \frac{9.1\times6\times3\times10^{10-31+19}}{1.6}$$

$$= 102.375\times10^{-2}$$

$$= 1.02\ \text{eV} \approx 1\ \text{eV}$$

$$E_{Ph} = \frac{1240}{\lambda}$$

$$E_{Ph} = \frac{1240}{400}\,\text{nm} = 3.1\ \text{eV}$$

$$E_{Ph} = \phi + \text{K.E}$$

$$\therefore \phi = 3.1 - 1 = 2.1\ \text{eV}$$

Q16 **Photoelectric emission is observed from a surface for frequencies ν_1 and ν_2 of incident radiations ($\nu_1 > \nu_2$). If the maximum kinetic energy of photoelectrons in the two cases are in the ratio of 2:1 respectively then threshold frequency ν_0 is given by**

(A) $\frac{2\nu_1 - \nu_2}{2-1}$ **(B)** $\frac{\nu_1 - \nu_2}{2-1}$ **(C)** $\frac{2\nu_2 - \nu_1}{2-1}$ **(D)** $\frac{\nu_2 - \nu_1}{2}$

A16 **(C)**

We know that,

$$E_{Ph} = \phi + KE_{max}$$

$$\Rightarrow \quad KE_{max} = h\nu - h\nu_0$$

$$K.E_1 = h\nu_1 - h\nu_0 \quad \ldots (1)$$

$$KE_2 = h\nu_2 - h\nu_0 \quad \ldots (2)$$

$$\frac{K.E_1}{KE_2} = \frac{2}{1} = \frac{h(\nu_1 - \nu_0)}{h(\nu_2 - \nu_0)}$$

$$2(\nu_2 - \nu_0) = (\nu_1 - \nu_0)$$

$$2\nu_2 - 2\nu_0 = \nu_1 - \nu_0$$

$$2\nu_2 - \nu_1 = -\nu_0 + 2\nu_0$$

$$\Rightarrow \quad \nu_0 = \left(\frac{2\nu_2 - \nu_1}{2-1}\right)$$

Q17 **In photoelectric effect, if a curve of $K.E_{max}$, of ejected photoelectron is plotted with respect to frequency of incident radiation (x–axis) on a metal surface of work function (ϕ) then slope of curve is**

(A) ϕ **(B) h** **(C) h/e** **(D) h.e**

A17 **(B)**

Given K.E on y–axis and ν on x – axis in a curve

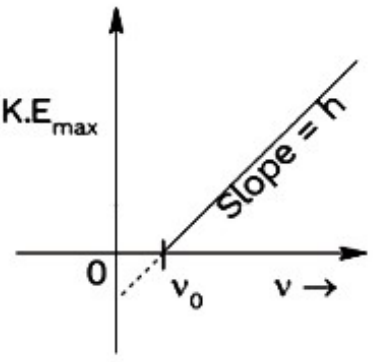
$$KE_{max} = h\nu - \phi$$

$$y = hx - \phi \quad \ldots(1)$$

General equation of line is $y = mx + c$ ...(2)

On comparing equation (1) and (2) we get

Slope, $\boxed{m = h}$ and $c = (-\phi)$



Q18 **A 64 Watt bulb emits monochromatic radiations of 310 nm, emitted photons from bulb strike on a metal surface of work function 4.7 eV. Then the number of photoelectrons emitted per second will be**

(A) 0 **(B) 10^{20}** **(C) 16×10^{19}** **(D) 4×10^{19}**

A18 **(A)**

Given: Work function ϕ = 4.7 eV

Wavelength of incident radiation = 310 nm

Power = 64 watt

$$E_{photon} = \frac{1240}{\lambda}\ eV$$

$$= \frac{1240}{310}\ eV$$

$$= 4\ eV$$

Since, $E_{ph} < \phi$, so there will be no ejection

$\therefore$ number of photoelectrons = 0

Q19 **A metal is exposed to a light source rated as 30 Watt is emitting radiation of wavelength 4000 Å then calculate:**

i) Number of photoelectrons per second.

ii) Photocurrent

A19 Given, Power = 30 Watt

Wavelength = 4000 Å

$$\text{Power} = \frac{\text{Energy}}{\text{Time}}\ \frac{J}{s}$$

Let time = 1 second and no. of photoelectrons emitted in 1 sec. = N

$$30 = \frac{N\left(\frac{hc}{\lambda}\right)}{1sec}$$

$$\Rightarrow \quad N = \frac{30 \times 4000 \times 10^{-10}}{6.6 \times 10^{-34} \times 3 \times 10^{8}}$$

$$N = \frac{4000 \times 10 \times 10^{-10+34-8}}{6.6}$$

$$N = \frac{40 \times 10^{-10+34-8+3}}{6.6}$$

$N = 6.06 \times 10^{19}$

Number of photon = no. of photoelectrons

$= 6.06 \times 10^{19}$

We known that, $I = q/t$

Magnitude of photocurrent $= \left(\frac{N_e}{t}\right) \times q_e$

$= 6.06 \times 10^{19} \times 1.6 \times 10^{-19} = 9.696$ ampere

Q20 **In an experiment of photoelectric effect a graph of $K.E_{max}$ of ejected photo-electron V/s frequency of incident light is plotted.**
The threshold wavelength (λ_0) of metal is
(A) 310 nm **(B) 400 nm** **(C) 4000 nm** **(D) 3100 nm**

A20 **(B)**

$(0, -3.1)$

We know that

$K.E = h\nu - \phi$

i.e. slope = h

Intercept (c) $= -\phi = -3.1$ eV

$$\boxed{\phi = 3.1 \text{ eV}}$$

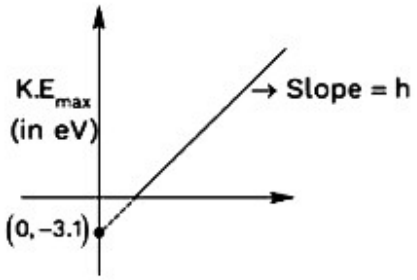
$$\phi = \frac{1240}{\lambda_0}$$

$$3.1 \text{ eV} = \frac{1240}{\lambda_0}$$

$$\lambda_0 = \frac{1240}{3.1}$$

$$\lambda_0 = \frac{1240}{31} \times 10$$

$$\lambda_0 = 400 \text{ nm}$$



Intensity of radiation

It is defined as the incident energy of radiation per unit time, per unit surface area.

$$\text{i.e. } \boxed{I = \frac{\text{Energy of radiation}}{\text{time x surface area}}}$$

According to classical physics, More intensity means more energy.

So with increase in intensity, the $K.E_{max}$ of photoelectron should increase but practically $K.E_{max}$ does not change. This also proves that a body absorbs energy in quantised or discrete manner.

But applying Planck's quantum theory

$$\boxed{I = \frac{(N.h\nu)}{t.A}}$$

So, for a metal plate exposed to a particular radiation.

i.e. ν = constant

A = constant

$$\Rightarrow I \propto \left(\frac{N}{t}\right)$$

So, More intensity $\Rightarrow$ More no of photons $\Rightarrow$ More ejection of photoelectrons $\Rightarrow$ photocurrent increases.

Q21 **In photoelectric experiment performed by H.Hertz as intensity of incident light on metal surface increases (For constant frequency):**

(A) K.E_{max} of photoelectron increases

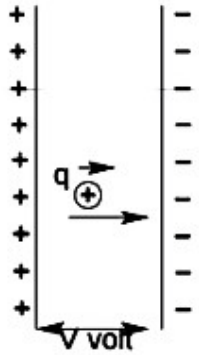
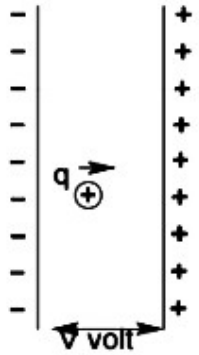
(B) Number of ejected photoelectrons increases

(C) Magnitude of photocurrent decreases

(D) K.$E._{max}$ of photoelectron decreases

A21 **(B)**

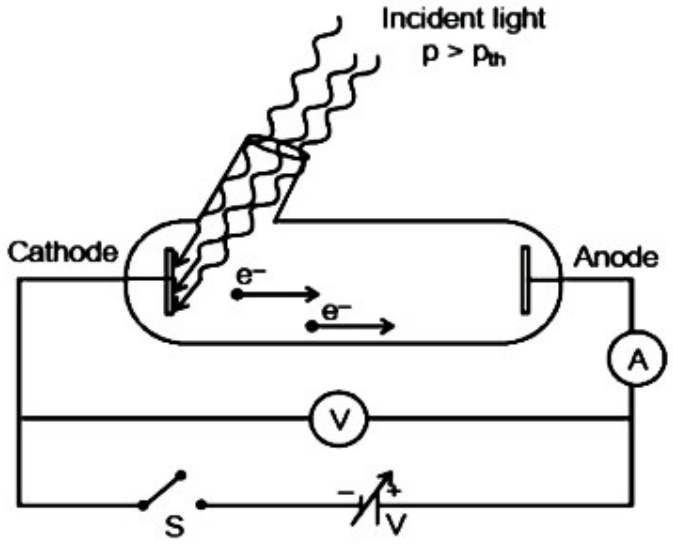
Note: If a charged particle moves in an electric filed:

(I) IF ACCELERATING POTENTIAL	RETARDING POTENTIAL
	
The increases in K.E=q×V	The decreases in K.E=q×V

Stopping potential (V_0)

The minimum potential which must be applied to stop the movement of photoelectron in the tube.

Means it is a retarding potential.



So, decrease in K.E of photoelectron = (e)×V_0

To completely stop the photoelectron,

Decrease in K.E of Photoelectron = $K.E_{max}$

$eV_0 = K.E_{max}$

$$\Rightarrow \boxed{V_0 = \frac{K.E_{max}}{e}}$$

Q22 **A source of electromagnetic radiation emits radiation of wavelength 1550 Å. Emitted radiation strikes on a metal surface of work function 4 eV then stopping potential of photoelectron is**

(A) 1V **(B) 2V** **(C) 3V** **(D) 4V**

A22 **(D)**

Given, wavelength = 1550 Å and work function ϕ = 4 eV

We known that $E_{Photon} = \frac{1240}{\lambda\,(\text{in nm})}$

$$= \frac{1240}{155}\ \text{eV}$$

$$= 8\,\text{eV}$$

$$K.E_{max} = E_{Ph} - \phi$$

$$= 8 - 4 = 4\ \text{eV}$$

$$V_0 = \frac{K.E_{max}}{e} = \frac{4\,\text{eV}}{e} = 4\,\text{volts}$$

Graphs

LESSON-8

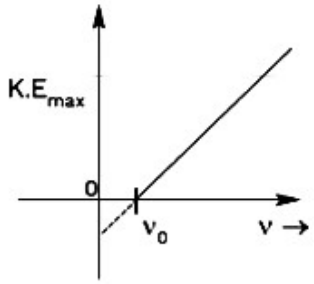
- GRAPHS OF PHOTOELECTRIC EFFECT
- BOHR'S MODEL
- DERIVATIONS FOR BOHR'S MODEL

1. $K.E_{max}$ vs ν

   $K.E_{max} = E_{Ph} - \phi$

   $K.E_{max} = h\nu - \phi$ ...(1)

   $y = mx + c$ ...(2)



On comparing equation (1) & (2) we get

Slope (m) = $h > 0$

Intercept (c) = $(-\phi) < 0$

2. V_0 vs ν_0

$$V_0 = \frac{K.E_{max}}{e}$$

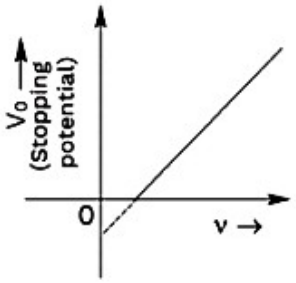
$$V_0 = \frac{h\nu - \phi}{e}$$

On comparing with $y = mx + c$, we get $y = V_0$

$$y = \frac{h}{e}x - \frac{\phi}{e}$$

So slope (m) $= \frac{h}{e}$

Intercept (c) $= \left(\frac{-\phi}{e}\right)$



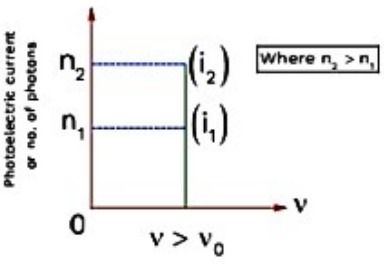
3. Photocurrent vs ν

$i \propto \nu^0$

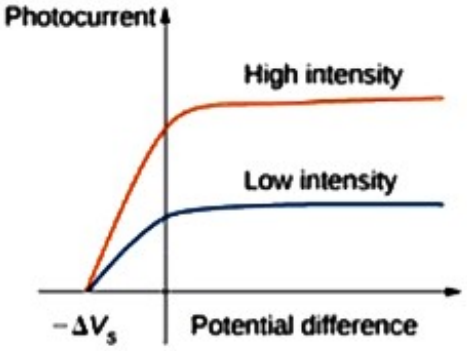
(independent of frequency)

$\Rightarrow$ i = constant

y



4. Photocurrent vs Potential difference



Note: Photoelectric effect supports particle nature of electromagnetic radiation.

Line spectrum of atoms

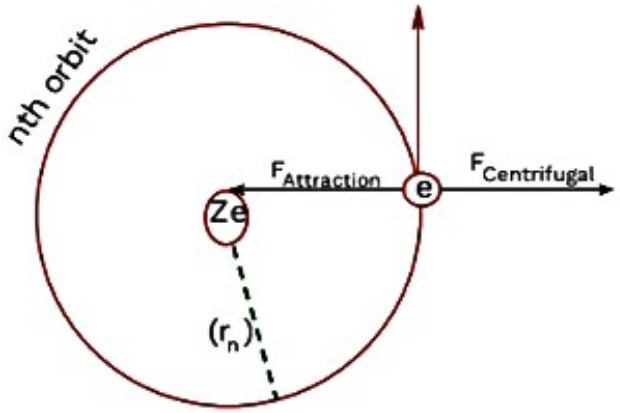
i) When atomic gas is heated, it gets energy and then emits electromagnetic radiations (EMR).
ii) When these radiations are traced on a photo graphic plate then only few well defined lines are obtained.
iii) Based upon concepts of Planck's quantum theory and classical physics, Bohr explained the atomic spectrum of H using his atomic model.

Bohr's Model

This model is applicable for H or H like species (Unielectronic species only).
e.g. H, He^{+}, Li^{2+}, Be^{3+} etc.
But for He, Li^{+}, Li, Be^{2+} etc, It is not applicable.

Important postulates:

The atom has a very small nucleus containing all protons and neutrons and this nucleus is present at the centre of the atom.
Electrons revolve around the nucleus in well-defined circular paths which are called "orbits"
Electrostatic attraction force between nucleus and the electron is balanced by the centrifugal force on the electron due to circular motion.



$$F_{attraction} = F_{centrifugal}$$

$$\frac{Kq_1q_2}{r_n^2} = \frac{mv_n^2}{r_n}$$

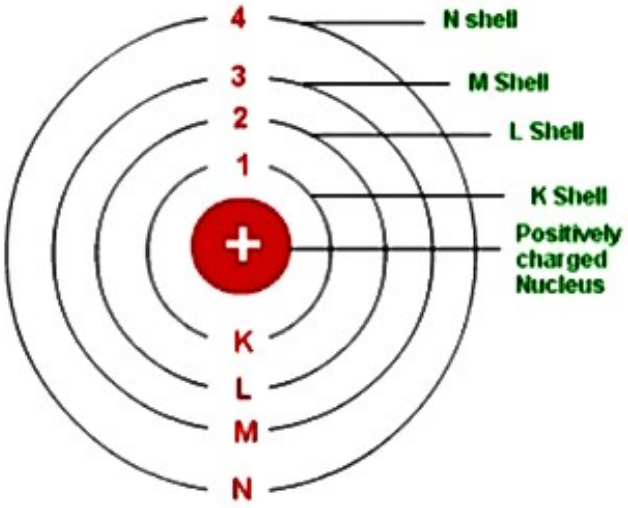
$$\Rightarrow \frac{K(Ze)(e)}{r_n} = mv_n^2$$

Electron can revolve only in those orbits for which the angular momentum of electron is a natural number multiple of $\frac{h}{2\pi}$.

Due to circular motion, angular momentum = mv_nr_n

i.e. $\boxed{mv_nr_n = \frac{nh}{2\pi}}$

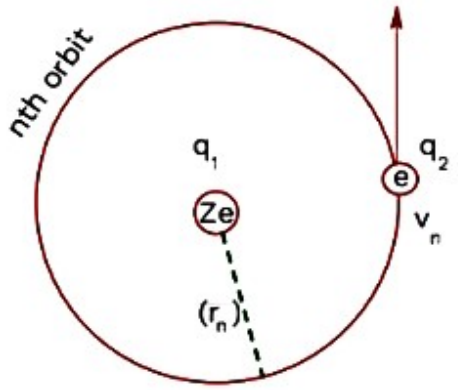
Where 'n' is natural number (1,2,3,4, ...)
and 'h' is Planck's constant.
$h = 6.626\times10^{-34}$ Js
Since, the net force on the electron is zero so the energy of electron in an orbit remains constant and hence these orbits are also known as "energy shells".
These orbits or shells are numbered as 1,2,3,4, ... Corresponding to the valence of 'n' which are also named as K, L, M, N ..... shells from nucleus outwards.



Electron moves from a lower orbit to a higher orbit by absorbing certain amount of energy as EMR.
Electron moves from higher orbit to a lower orbit by releasing certain amount of energy as EMR.

Derivations for Bohr's model
Applicable only and only for unielectronic species

For the electron present in nth orbit:



From postulates

$$F_{attraction} = F_{centrifugal}$$

$$\Rightarrow \quad \frac{Kq_1q_2}{r_n^2} = \frac{mv_n^2}{r_n}$$

$$\Rightarrow \quad \frac{K(Ze)(e)}{r_n} = mv_n^2 \qquad ...(1)$$

$$\Rightarrow \quad mv_nr_n = \frac{nh}{2\pi} \qquad ...(2)$$

Radius of nth orbit (r_n)
From equation (2)

$$v_n = \frac{nh}{2\pi mr_n}$$

putting the value of v_n in eq. (1),

$$\frac{K.Z.e^2}{r_n} = m\left(\frac{nh}{2\pi mr_n}\right)^2$$

$$\frac{K.Z.e^2}{r_n} = \frac{mn^2h^2}{4\pi^2m^2r_n^2}$$

$$r_n = \frac{h^2}{4\pi^2K.me^2}\left(\frac{n^2}{Z}\right)$$

$$\Rightarrow \boxed{r_n = 0.529\left(\frac{n^2}{Z}\right)\text{Å}}$$

Where 'n' is orbit number
Z is atomic number
For a particular species (Z=constant)
$r_n \propto n^2$
$\Rightarrow$ If 'n' increases then 'r' also increases of orbits.
i.e. $r_1<r_2<r_3<r_4$ ....

Speed of electron in n^{th} orbit (v_n)

Put $r_n = \frac{n^2h^2}{4\pi^2Kme^2Z}$ put in eq. (2)

$$mv_n\left(\frac{n^2h^2}{4\pi^2Kme^2Z}\right) = \frac{nh}{2\pi}$$

$$v_n = \frac{2\pi K.e^2\left(\frac{Z}{n}\right)}{h}$$

$\Rightarrow$ $\boxed{v_n = 2.18\times10^6\left(\frac{Z}{n}\right)}$ m/s

For particular species (Z= constant)

$\boxed{v_n \propto \frac{1}{n}}$

$\Rightarrow$ If 'n' increases then v decreases of electron.
$\Rightarrow$ $v_1 > v_2 > v_3 > v_4$ ...
$\Rightarrow$ $KE_1 > KE_2 > KE_3 > KE_4$ ...

Practice questions for Bohr's Theory

CONTENT

- PRACTICE QUESTIONS FOR BOHR'S THEORY
- ENERGY OF ELECTRON IN n^{th} ORBIT (TOTAL, KINETIC, POTENTIAL)
- APPLICATIONS OF BOHR'S MODEL
  1. Time period
  2. Frequency of electron revolution (F)
  3. Absorption / Emission of energy

Q23 **For H–atom, calculate**
(i) Radius of 3rd orbit **(ii) Speed of electron in 2nd orbit**
(iii) Radius of 4th – shell

A23 Using $r_n = 0.529\left(\frac{n^2}{Z}\right)$ Å

i) Radius of 3rd orbit
n=3, Z=1

$$r_{3,H} = 0.529 \times \frac{3^2}{1} = 4.761 \text{ Å}$$

ii) Speed of electron in 2nd orbit

Using $v_n = 2.18 \times 10^6 \left(\frac{Z}{n}\right)$ m / s

$$v_{2,H} = 2.18 \times \frac{1}{2} \times 10^6 = 1.09 \times 10^6 \text{ m / s}$$

iii) Radius of 4th – shell
n=4

$$r_{4,H} = 0.529 \times \frac{4^2}{1} = 8.464 \text{ Å}$$

Q24 **Calculate speed of electron in 1st Bohr orbit of He^+**

A24 Atomic no. of $He^+ = 2$

$$v_n = 2.18\times10^6 \left(\frac{Z}{n}\right) m/s = 2.18 \times 10^6 \times \frac{2}{1} = 4.36 \times 10^6 \ m/s$$

Q25 **Calculate radius of L–shell in Li^{2+}.**

A25 Given; n=2 (∵ L shell)

For Li^{2+}, Z=3

$$r_n = 0.529 \times \frac{n^2}{Z} Å = 0.529 \times \frac{(2)^2}{3} = 0.705 Å$$

Q26 **In H–atom, calculate the ratio of radii of 1st and 4th orbits.**

A26 Given, H–atom

For H–atom , Z =1

$n_1 = 1$, and $n_2 = 4$ (Given)

Using, $r_n \propto n^2$ ... only for same element

Ratio of radii of 1st and 4th orbits,

$$\frac{r_1}{r_4} = \left(\frac{1}{4}\right)^2 = \frac{1}{16}$$

Q27 **For H–atom, calculate the ratio of speeds of electron is K–shell to M–shell**

A27 For K shell, n=1 and for M–shell, n=3
We know that

$$v \propto \frac{1}{n}, \quad \text{only for same element}$$

$$\frac{v_1}{v_2} = \frac{\frac{1}{n_1}}{\frac{1}{n_2}} = \frac{n_2}{n_1}$$

$$\boxed{\frac{v_1}{v_2} = \frac{3}{1}}$$

Q28 **Calculate the ratio of radius of 2nd orbit in He^+ to the radius of 3rd orbit in Be^{3+}.**

A28 We know that $r_n \propto \frac{n^2}{Z}$

Ratio of radius of 2nd orbit in He^+ to the radius of 3rd orbit in Be^{3+}

$$\text{i.e.} \Rightarrow \frac{r_{2,He^+}}{r_{3,Be^{+3}}} = \frac{\left(\frac{2^2}{2}\right)}{\left(\frac{3^2}{4}\right)} = \frac{\frac{4}{2}}{\frac{9}{4}} = \frac{4}{2} \times \frac{4}{9} = \frac{8}{9}$$

Q29 **Calculate the ratio of speeds of electron in 3rd orbit of Li^{2+} to that in N–shell of H.**

A29 We know $v_n \propto \left(\frac{Z}{n}\right)$

Ratio of speeds of electrons in 3rd orbit of Li^{2+} to that of N–shell of H.

i.e. $\frac{V_{3,Li^{+2}}}{V_{4,H}} = \frac{\left(\frac{3}{3}\right)}{\left(\frac{1}{4}\right)} = \frac{4}{1}$

Q30 **In H–atom, for the electron revolving in n^{th} orbit if $r_n = 9r_1$ then calculate speed of this electron.**

A30 Given $r_n = 9r_1$

$$\Rightarrow \quad \frac{r_n}{r_1} = \frac{9}{1}$$

We know that

$$r_n \propto n^2$$

$$\frac{r_n}{r_1} = \frac{n^2}{1^2} = \frac{9}{1} = \frac{n^2}{1^2} \Rightarrow n^2 = 9$$

$$\boxed{n = 3}$$

So, $v = 2.18 \times 10^6 \left(\frac{Z}{n}\right) = 2.18 \times 10^6 \left(\frac{1}{3}\right)$

$$\boxed{v_{3,H} = 0.72 \times 10^6 \text{ m/s}}$$

Energy of electron in n^{th} orbit :

There are two types of energy for the electron as following:

i) **Due to movement of e^- (K.E)**

$$K.E = \frac{1}{2}mv^2$$

As for e– in unielectronic system:

$$\Rightarrow \quad \frac{K.Ze^2}{r_n} = mv_n^2$$

$$\boxed{K.E_n = \frac{K.Z.e^2}{2r_n}}$$

ii) **Due to electrostatic attraction of nucleus (PE)**

$$P.E_n = \frac{Kq_1q_2}{r_n} = \frac{K(+Ze)(-e)}{r_n}$$

$$\boxed{P.E_n = \frac{-K.Ze^2}{r_n}}$$

Total energy of electron (E_n)

$$E_n = K.E_n + P.E_n = \left(\frac{KZe^2}{2r_n}\right) + \left(\frac{-KZe^2}{r_n}\right)$$

$$\boxed{E_n = -\frac{KZe^2}{2r_n}}$$

where, $\boxed{r_n = 0.529 \times \frac{n^2}{Z} \times 10^{-10} m}$

$$\Rightarrow \quad \boxed{E_n = -13.6\left(\frac{Z^2}{n^2}\right)\frac{eV}{atom}}$$

Or

$$\boxed{E_n = -2.18 \times 10^{-18}\left(\frac{Z^2}{n^2}\right)\frac{J}{atom}}$$

Note: Due to convention, the total energy of a free electron at rest is considered zero. Since the electron inside an atom is more stable than the free electron.
So, its energy must be lesser as compared to the energy of free electron.
i.e. energy of e– inside the atom is taken negative

Note : i) $\boxed{K.E = -E_n}$

ii) $\boxed{P.E = 2E_n}$

Where, $E_n = -13.6\left(\frac{Z^2}{n^2}\right)$ eV

For particular species (Z=constant)

$$\boxed{(-E_n) \propto \frac{1}{n^2}}$$

as 'n' increases $\Rightarrow -E_n$ decreases $\Rightarrow E_n$ increases
As 'n' decreases $\Rightarrow -E_n$ increases $\Rightarrow E_n$ decreases
i.e. $E_1 < E_2 < E_3 < E_4$ .....
Electron is most stable in 1st orbit and hence 1st orbit is also known as ground state.
For H–atom (Z=1)

$$E_1 = -13.6 \times \left(\frac{1^2}{1^2}\right) = -13.6 \text{ eV}$$

$$E_2 = -13.6\left(\frac{1}{2^2}\right) = -3.4 \text{ eV}$$

$$E_3 = -13.6\left(\frac{1}{9}\right) = -1.51 \text{ eV}$$

$$E_4 = -13.6\left(\frac{1}{16}\right) = -0.85 \text{ eV}$$

$$\boxed{(E_n)_Z = (E_n)_H . Z^2}$$

Q31 **Calculate energy of electron in L–shell of Li^{2+}.**

A31 Given for Li^{2+}, Z=3 and shell, n=2
We know that

$$E_n = -13.6 \times \frac{Z^2}{n^2} = -13.6 \times \frac{3^2}{2^2} = -30.6 \text{ eV}$$

or

$$(E_2)_{Li^{2+}} = (E_2)_H \times Z^2 = (-3.4) \times 3^2 = -30.6 \text{ eV}$$

Q32 **Calculate kinetic energy of electron present in 2nd orbit of He^+.**

A32 Given n=2, Z=2, (He^+)
We know that $K.E_n = -E_n$

Where $E_n = -13.6 \times \frac{Z^2}{n^2} = -13.6 \times \frac{2^2}{2^2} = (-13.6) \text{ eV}$

$$\boxed{K.E_{2,He^+} = +13.6 \text{ eV}}$$

Q33 **In Be^{3+} calculate ratio of energies of electron in 3rd orbit to the 4th orbit.**

A33 For Be^{3+}, Z=4

We know that $-(E_n) \propto \frac{1}{n^2}$

A.T.Q $$\frac{E_3}{E_4} = \frac{-\left(\frac{1}{3}\right)^2}{-\left(\frac{1}{4}\right)^2} = \frac{16}{9}$$

Q34 **Calculate ratio of energies of electron in K–shell of He^+ to that in 2nd orbit of H.**

A34 We know that $(E_n) \propto \frac{Z^2}{n^2}$

$$\frac{E_{1,He^+}}{E_{2,H}} = \frac{\left(\frac{2^2}{1^2}\right)}{\left(\frac{1^2}{2^2}\right)} = \frac{16}{1}$$

Q35 **In Li^{2+} for the electron revolving in n^{th} orbit if $v_1 = 2v_n$, then calculate energy of this electron.**

A35 We know that for a particular species $v \propto \frac{1}{n}$

$$\frac{v_1}{v_n} = \frac{\left(\frac{1}{1}\right)}{\left(\frac{1}{n}\right)} = \frac{n}{1} = \frac{2}{1} \Rightarrow n = 2$$

$$E_{2,Li^{2+}} = (-3.4) \times 2^2 = -13.6 \text{ eV}$$

Applications of Bohr's model

1. Time period

The time taken by the electron in one complete revolution of the orbit.

$\therefore$ time $= \dfrac{\text{distance travelled}}{\text{speed}}$

$$T_n = \frac{2\pi r_n}{v_n}$$

$$T_n = 1.52 \times 10^{-16} \times \frac{n^3}{Z^2} \text{ sec}$$

(for 1 – revolution)

Also, $T_n \propto \dfrac{\left(\frac{n^2}{Z}\right)}{\left(\frac{Z}{n}\right)}$

$$\boxed{T_n \propto \left(\frac{n^3}{Z^2}\right)}$$

2. Frequency of electron revolution (F)

The number of revolutions made by electron in one second.

In T seconds, there will be 1 revolution.

In 1 second, there $\left(\dfrac{1}{T} \times 1\right)$ revolutions

i.e. $\boxed{F_n = \dfrac{1}{T_n} = \dfrac{v_n}{2\pi r_n}} = 0.657 \times 10^{16} \times \dfrac{Z^2}{n^3} \text{ s}^{-1}$

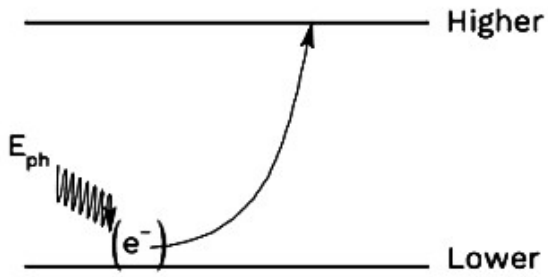
$$\boxed{F_n \propto \frac{Z^2}{n^3}}$$

3. Absorption / Emission of energy

Absorption

⇒ When the electron absorbs a photon of certain fixed energy, it moves from its orbit to a higher orbit.

- The energy of quantum must be exactly equal to the exact energy difference between these two orbits.



For lower to higher

E_{ph} = ΔE(exactly)

= $E_H - E_L$

This process is known as excitation and hence energy shells are also named as following:

n=5 _____ 4th excited state

n=4 _____ 3rd excited state

n=3 _____ 2nd excited state

n=2 _____ 1st excited state

n=1 _____ (e^-) _____ Ground state

Separation Energy

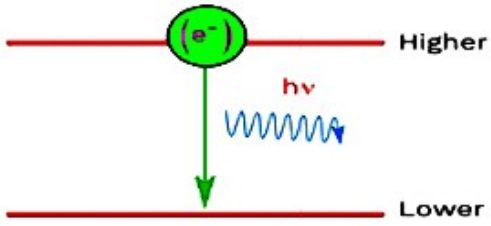
LESSON–10

- EMISSION
- WAVELENGTH OF PHOTON
- IONIZATION ENERGY OF UNIELECTRONIC SPECIES.

EMISSION

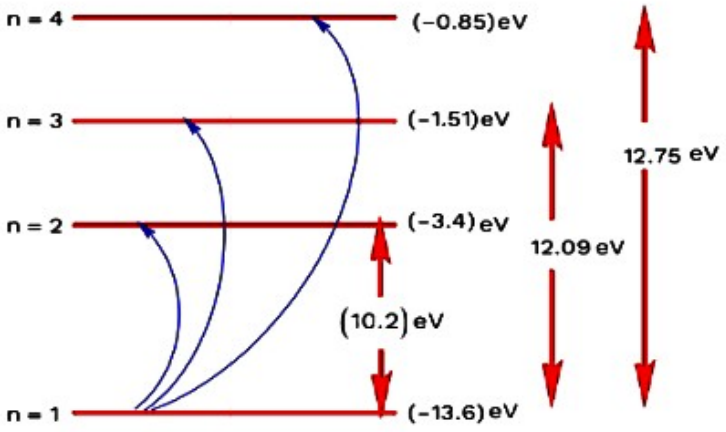
When an electron jumps from a higher orbit to a lower orbit, a quantum is emitted.

Energy of this emitted quantum is exactly equal to the energy difference between both orbits.



$$h\nu = E_{Photon} = \Delta E = (E_H - E_L)$$

e.g For H–atom



i.e. For any transition,

$$\boxed{E_{Ph} = E_f - E_i}$$

Q36 **Ground state electron of H–atom is exposed to a radiation of 10.3 eV. Predict what will happen.**

A36 Ground state electron energy = –13.6 eV

Energy of electron = $\left[-13.6 + 10.3\right]$ eV = –3.3 eV

So, there is no energy shell corresponding to –3.3 eV. So this photon will not be absorbed by the electron.

Q37 **In He^+ calculate the energy of photon required to move the ground state electron to 2nd excited state.**

A37 For He^+ , Z = 1

$n_1 = 1 \rightarrow n_2 = 3$

$$E_{Photon} = (E_3 - E_1)_{He^+} = -13.6 \times \frac{Z^2}{n_2^2} - \left(-13.6 \times \frac{Z^2}{n_1^2}\right) \qquad \left[\therefore E_n = -13.6 \times \frac{Z^2}{n^2}\right]$$

$$= -13.6 \times \frac{2^2}{3^2} - \left(-13.6 \times \frac{2^2}{1^2}\right)$$

$$= -13.6 \times \frac{4}{9} + 13.6 \times 4$$

$$= 13.6\left(4 - \frac{4}{9}\right)$$

$$= \frac{13.6 \times 32}{9}$$

$$= 48.3 \text{ eV}$$

Or

$$E_{Ph} = \left(E_3 - E_1\right)_H \times Z^2 = 12.09 \times 2^2 = 48.36 \text{ eV}$$

Q3 **Calculate energy of photon emitted when electron present in N–Shell jump to 1st excited state in H–atom.**

A3 **Sol.:** $(n_1 = 4) \rightarrow (n_2 = 2)$

$E_{Ph} = E_4 - E_2 = -0.85 - (-3.4) = 2.55$ eV

4. Wavelength of Photon

In only electronic transition,

$E_{Ph} = (E_H - E_L)$

$$\frac{hc}{\lambda} = \left(-21.8 \times 10^{-19}\right)\left(\frac{1}{n_H^2} - \frac{1}{n_L^2}\right).Z^2 \text{ Joule}$$

$$\frac{1}{\lambda} = \frac{\left(+21.8 \times 10^{-19}\right)}{\left(19.878 \times 10^{-26}\right)}\left(\frac{1}{n_L^2} - \frac{1}{n_H^2}\right).Z^2$$

Here, $\left(\frac{21.8 \times 10^{-19}}{19.878 \times 10^{-26}}\right) = 109677\,cm^{-1}$

$\approx 109700\,cm^{-1} = R_H$(Rydberg constant)

$$\Rightarrow \frac{1}{\lambda} = R_H\left(\frac{1}{n_L^2} - \frac{1}{n_H^2}\right).Z^2$$

Where n_L = lower orbit number

n_H = higher orbit number

If $\boxed{R_H = 109700 cm^{-1}}$

$\boxed{\lambda \rightarrow cm}$

Also, $\boxed{\left(\frac{1}{R_H}\right) \simeq 912Å}$

$\Rightarrow \lambda \rightarrow Å$

Q38 **The wavelength of the radiation emitted when in a hydrogen atom electron falls from infinity to stationary state 1, would be:**
(Rydberg constant = $1.097\times10^{7}\ m^{-1}$)
(A) 91.2 nm **(B) 192 nm** **(C) 406 nm** **(D) 9.1×10^{-8} nm**

A38 **A**

$(n_H = \infty) \rightarrow (n_L = 1)$

We know that $\Rightarrow \frac{1}{\lambda} = R_H\left(\frac{1}{n_L^2} - \frac{1}{n_H^2}\right).Z^2$

$\Rightarrow \quad \frac{1}{\lambda} = R_H\left(\frac{1}{1^2} - \frac{1}{\infty^2}\right).1^2$

$\Rightarrow \frac{1}{\lambda} = R_H$

$\Rightarrow \lambda = \left(\frac{1}{R_H}\right)$

$\Rightarrow \lambda = 912\ Å$

$\Rightarrow \lambda = 91.2\ nm$

Q39 **Calculate wavelength of photon required to excite the electron from L–shell to 3rd excited state in He^+.**

A39 L shell $\rightarrow$ 3rd excited state

$(n_L=2)$ to $(n_H=4)$

He^+ $(Z=2)$

We known that $\frac{1}{\lambda} = R_H\left(\frac{1}{n_L^2} - \frac{1}{n_H^2}\right).Z^2$

$\frac{1}{\lambda} = R_H\left(\frac{1}{2^2} - \frac{1}{4^2}\right).2^2$

$\frac{1}{\lambda} = R_H\left(\frac{3}{4}\right)$

$\lambda = \frac{4}{3} \times \left(\frac{1}{R_H}\right)$

$\lambda = \frac{4}{3} \times 912$

$\lambda = 1216\ Å$

5. Ionization energy of unielectronic species.

The minimum energy required to remove the ground state electron out of the atom.

i.e. $n=1 \rightarrow n=\infty$

$\therefore \quad E_{Ph} = E_H - E_L$

$\Rightarrow \quad I.E = E_\infty - E_1$

$I.E = 0 - E_1$

$$\boxed{I.E = (-E_1)}$$

e.g. For H–atom = +(13.6) eV.

6. Separation energy:

The minimum energy required by an excited state electron to move out of the atom.

i.e. $n_H = \infty$

$n_L = 2,3,4,5 \ldots\ldots\ldots \quad (n_L > 1)$

$E_{Ph} = E_H - E_L$

$S.E = E_\infty - E_L$

$S.E = 0 - E_L$

$$\boxed{S.E = (-E_L)} \quad n_L > 1$$

Q40 **Calculate I.E of Li^{2+} and Be^{3+}.**

A40

Li^{2+}	Be^{3+}
$E_n = -13.6 \times \frac{Z^2}{n^2}$ $E_1 = -13.6 \times \frac{3^2}{1^2}$ eV $E_1 = -122.4$ eV I.E $= -E_1$ I.E = 122.4 eV	$E_1 = -13.6 \times \frac{4^2}{1^2}$ $E_1 = -217.6$ eV I.E = 217.6 eV

Q41 **Calculate I.E of H in kJ/mol.**

A41

$E_1 = -13.6$ eV/atom

I.E = 13.6 eV/atom

I.E = $13.6 \times 1.6 \times 10^{-19}$ J/atom

$$\text{I.E} = \frac{13.6 \times 1.6 \times 10^{-19}}{1000} \text{kJ / atom}$$

$$\text{I.E} = \frac{13.6 \times 1.6 \times 10^{-19}}{10^3} \times 6.02 \times 10^{23} \text{ kJ/mole}$$

I.E = $130.995 \times 10^{-19+23-3}$ kJ/mole

I.E = 130.995×10^{1} kJ/mole

I.E = 1309.95 kJ/mol

Q42 **Ionization energy of a unielectronic species is 122.4 eV/atom, then calculate radius of its third orbit.**

A42 Given:

$IE = 122.4$ eV

$IE = -(E_1)$

i.e. $E_1 = -122.4$ eV

$$-13.6 \times \frac{Z^2}{1^2} = -122.4$$

$\Rightarrow$ $Z = 3$

$\Rightarrow$ So, $r_{3,Li^{+2}} = 0.529 \times \frac{3^2}{3}$ Å

$r_{3,Li^{+2}} = 1.58 \times 10^{-10}$ m

Q43 **If a photon of 15 eV is provided to the ground state electron of H–atom. Predict what will happen.**

A43 $\because$ I.E = (13.6 eV)

I.E = minimum energy required by e– to come out of the atom

But, $E_{ph} = 15.$ eV

So, the e– will come out of the atom with K.E = 15–13.6 = 1.4 eV

Hydrogen Spectrum

LESSON–11

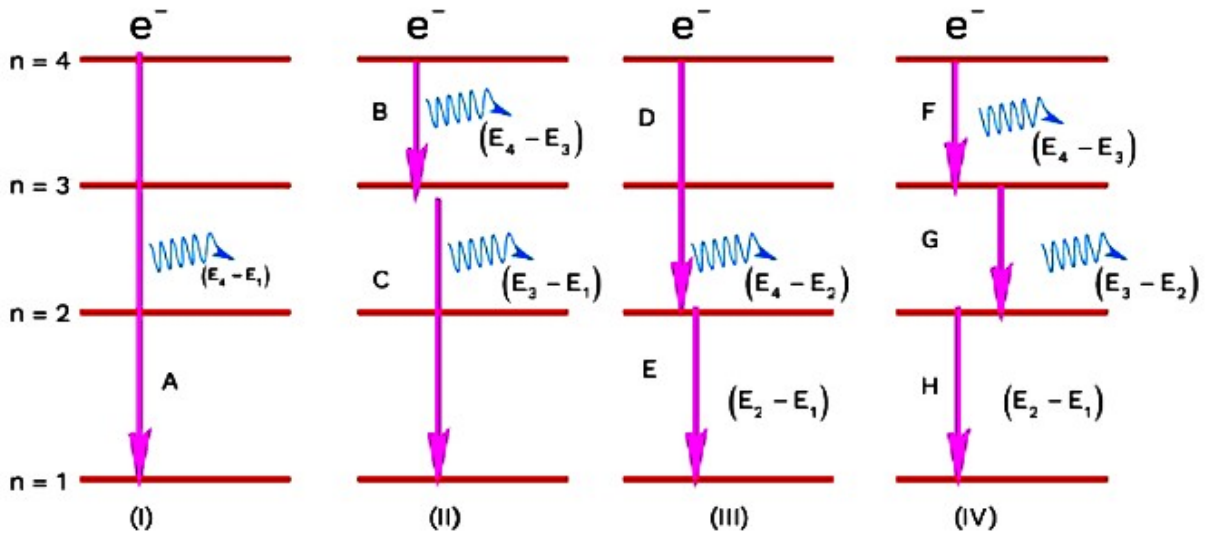
- HYDROGEN SPECTRUM
- SPECTRAL SERIES HYDROGEN
- WAVELENGTH OF ANY LINE
- TOTAL NUMBER OF UNIQUE SPECTRAL LINES

Line spectrum of hydrogen

When atomic gas is heated or electrically sparked, all atoms get excited, means all ground state electrons of atoms jump to higher energy level.

When these electrons return back to ground state, a large number of photons of different wavelengths are emitted due to transition of e– in all possible manner.

There can be four ways of arranging photons emitted if transition of e– is from n = 4 to n = 1.



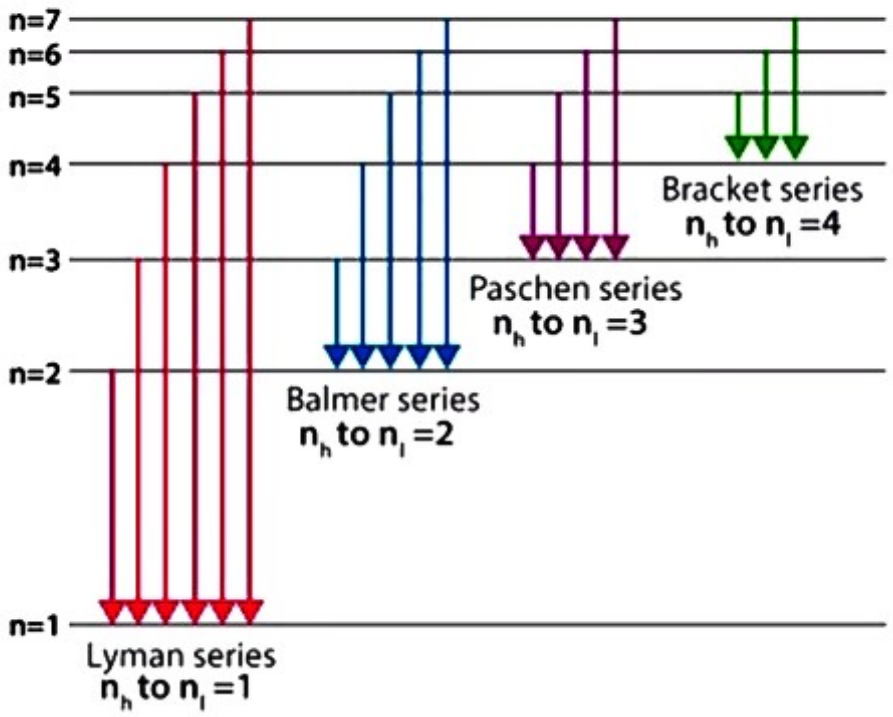
8 photons but unique = 6

In the four ways of representation of photons from the above figure, it can be seen that B & F, E & H are similar spectral lines. So, the net number of different types of spectral lines at any instance would be 6 only.

These wavelengths are grouped in certain different sets which are known as spectral series.

Spectral Series Hydrogen:
The six types of spectral series as follows:

Spectral series	n_L	n_H	Region of EMR (For H–only)
Lyman	1	2,3,4 ...∞	UV region
Balmer	2	3,4,5 ...∞	Visible
Paschen	3	4,5,6 ...∞	IR region
Brackett	4	5,6,7 ...∞	IR region
P Fund	5	6,7,8 ...∞	IR region
Humphrey	6	7,8,9 ...∞	Far IR region



Important notes

1. **Wavelength of any line**

$$\frac{1}{\lambda} = R_H\left(\frac{1}{n_L^2} - \frac{1}{n_H^2}\right)Z^2$$

2. **For a particular series: (n_L = fixed)**

1st line or α line $\Rightarrow (n_L+1) \xrightarrow{\text{to}} (n_L)$

2nd line or β line $\Rightarrow (n_L+2) \xrightarrow{\text{to}} (n_L)$

3rd line or γ line $\Rightarrow (n_L+3) \xrightarrow{\text{to}} (n_L)$

Last line or series limit $\Rightarrow \infty \xrightarrow{\text{to}} (n_L)$

3. **Line of longest wavelength**

$$\because \quad E_{ph} = \Delta E$$

$$\frac{hc}{\lambda} = \Delta E$$

$$\lambda \propto \frac{1}{\Delta E}$$

Longest wavelength $\Rightarrow$ minimum $\Delta E \Rightarrow$ 1st line

4. **Line of shortest wavelength:**
i.e. Max. $\Delta E \Rightarrow$ last line

5. **Total number of unique spectral lines**
When in an atomic gas sample an electron moves from n_H to n_L in all possible ways, then

$$\text{Total no. of unique spectral lines} = \frac{(n_H - n_L)(n_H - n_L + 1)}{2}$$

Lines of a particular series = $n_H - (n_L)_{series}$
e.g. $4 \rightarrow 1$

$$\text{Total no. of unique spectral lines} = \frac{(n_H - n_L)(n_H - n_L + 1)}{2}$$

$$= \frac{(4-1)(4-1+1)}{2}$$

$$= 6$$

Q44 **Calculate number of spectral lines observed when the electron present in 2nd excited state of H atom returns to ground state.**

(A) 1 **(B) 2** **(C) 3** **(D) 4**

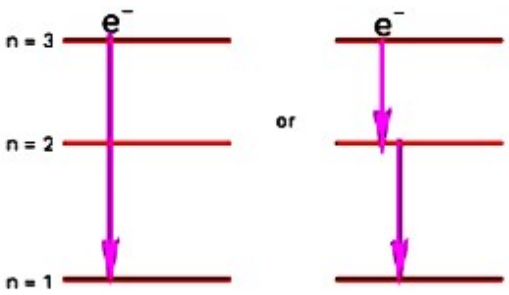
A44 **(C)**

$$\text{No. of spectral lines} = \frac{(n_H - n_L)(n_H - n_L + 1)}{2}$$

$$= \frac{(3-1)(3-1+1)}{2}$$

$$= \frac{3 \times 2}{2}$$

$$= 3$$



Given only H – atom

Minimum no of spectral lines observed = 1

Maximum no of spectral lines observed = 2

Q45 **Calculate wavelength for second line of Lyman series of He^+.**

A45 Given, Second line of Lyman series for He^+

i.e. $3 \rightarrow 1$

We know that

$$\frac{1}{\lambda} = R_H \left(\frac{1}{n_L^2} - \frac{1}{n_H^2} \right) Z^2$$

$$= R_H \left(\frac{1}{1^2} - \frac{1}{3^2} \right) Z^2$$

$$= R_H \left(\frac{8}{9} \right) \times 2^2$$

$$\frac{1}{\lambda} = R_H \frac{32}{9}$$

$$\lambda = \frac{9}{32 R_H} = \frac{9 \times 912}{32} \text{ Å}$$

Q46 **The shortest wavelength for He^+ in Balmer series is 'x', the longest wavelength in the Paschen series of Li^{2+} is:**

(A) $\frac{36x}{5}$ **(B)** $\frac{16x}{7}$ **(C)** $\frac{9x}{5}$ **(D)** $\frac{5x}{9}$

A46 **(B)**

$\infty \rightarrow$ **2 in** He^+	$4 \rightarrow 3$ **in** Li^{2+}
$\frac{1}{\lambda} = R_H\left(\frac{1}{n_L^2} - \frac{1}{n_H^2}\right)Z^2$	$\frac{1}{\lambda} = R_H\left(\frac{1}{n_L^2} - \frac{1}{n_H^2}\right)Z^2$
$\frac{1}{x} = R_H\left(\frac{1}{2^2} - \frac{1}{\infty^2}\right)2^2$	$\frac{1}{\lambda} = R_H\left(\frac{1}{3^2} - \frac{1}{4^2}\right)3^2$
$\frac{1}{x} = R_H\left(\frac{1}{4} - \frac{1}{0}\right) \times 4$	$\frac{1}{\lambda} = R_H\left(\frac{1}{9} - \frac{1}{16}\right) \times 9$
$\frac{1}{x} = R_H\left(\frac{1}{4} \times 4\right)$	$\frac{1}{\lambda} = R_H\left(\frac{16-9}{144}\right) \times 9$
$\boxed{\frac{1}{x} = R_H}$ ....(1)	$\frac{1}{\lambda} = R_H\left(\frac{7}{144}\right) \times 9$
	$\frac{1}{\lambda} = R_H\left(\frac{7}{16}\right)$ ...(2)

On dividing equation (1) by (2) we get

$$\frac{\frac{1}{x}}{\frac{1}{\lambda}} = \frac{R_H}{R_H\left(\frac{7}{16}\right)}$$

$$\frac{\lambda}{x} = \frac{16}{7}$$

$$\boxed{\lambda = \frac{16x}{7}}$$

$\therefore$ Longest wavelength in the Paschen series of Li^{2+} is $\frac{16x}{7}$

47 **The longest wavelength of He^+ ion of Paschen series is 'x', then shortest wavelength of Be^{3+} in Paschen series is (in terms of x):**

(A) $\frac{5}{36}x$ **(B)** $\frac{64}{7}x$ **(C)** $\frac{53}{8}x$ **(D)** $\frac{7}{64}x$

47 **(D)**

$4 \rightarrow 3$ in He^+	$\infty \rightarrow 3$ in Be^{3+}
$\frac{1}{\lambda} = R_H\left(\frac{1}{n_L^2} - \frac{1}{n_H^2}\right)Z^2$ $\frac{1}{x} = R_H\left(\frac{1}{3^2} - \frac{1}{4^2}\right)2^2$ $\frac{1}{x} = R_H\left(\frac{1}{9} - \frac{1}{16}\right) \times 4$ $\frac{1}{x} = R_H\left(\frac{16-9}{144}\right) \times 4$ $\frac{1}{x} = R_H\left(\frac{7}{144}\right) \times 4$ $\frac{1}{\lambda} = \frac{1}{x} = R_H\frac{7}{36}$ ....(1)	$\frac{1}{\lambda} = R_H\left(\frac{1}{3^2} - \frac{1}{\infty^2}\right) \times 4^2$ $\frac{1}{\lambda} = R_H\left(\frac{1}{9} - 0\right) \times 16$ $\frac{1}{\lambda} = R_H\left(\frac{16}{9}\right)$ ....(2)

On dividing equation (1) by (2), we get

$$\frac{\lambda}{x} = \frac{R_H\left(\frac{7}{36}\right)}{R_H\left(\frac{16}{9}\right)}$$

$$\frac{\lambda}{x} = \frac{7}{36} \times \frac{9}{16}$$

$$\boxed{\lambda = \frac{7}{64}x}$$

Q48 **The ratio of wavelength of photon corresponding to the α–line of Lyman series in H–atom and β–line of Balmer series in He^+ is:**

(A) 1:1 **(B) 1:2** **(C) 1:4** **(D) 3:16**

A48 **(A)**

α–line of Lyman series in H–atom	**β–line of Balmer series in He^+**
$\frac{1}{\lambda_1} = R_H\left(\frac{1}{1^2} - \frac{1}{2^2}\right) \times 1^2$ $\frac{1}{\lambda_1} = R_H\left(\frac{1}{1} - \frac{1}{4}\right) \times 1$ $\frac{1}{\lambda_1} = R_H\left(\frac{4-1}{4}\right) \times 1$ $\frac{1}{\lambda_1} = R_H \times \frac{3}{4} \times 1$ $\frac{1}{\lambda_1} = R_H\left(\frac{3}{4}\right)$ ...(1)	i.e. $4 \rightarrow 2$ $\frac{1}{\lambda} = R_H\left(\frac{1}{n_L^2} - \frac{1}{n_H^2}\right) Z^2$ $\frac{1}{\lambda_2} = R_H\left(\frac{1}{2^2} - \frac{1}{4^2}\right) 2^2$ $\frac{1}{\lambda_2} = R_H\left(\frac{1}{4} - \frac{1}{16}\right) \times 4$ $\frac{1}{\lambda_2} = R_H\left(\frac{4-1}{16}\right) \times 4$ $\frac{1}{\lambda_2} = R_H \times \frac{3}{16} \times 4$ $\frac{1}{\lambda_2} = R_H\left(\frac{3}{4}\right)$ ...(2)

On dividing equation (2) by (1), we get

$$\frac{\frac{1}{\lambda_2}}{\frac{1}{\lambda_1}} = \frac{R_H\left(\frac{3}{4}\right)}{R_H\left(\frac{3}{4}\right)}$$

$$\frac{\lambda_2}{\lambda_1} = \frac{1}{1}$$

$$\frac{\lambda_1}{\lambda_2} = 1:1$$

Drawback of Bohr's Model

LESSON-12

- DRAWBACK OF BOHR'S MODEL
- de–BROGLIE HYPOTHESIS
- PROBLEMS OF de–BROGLIE
- PROOF FOR BOHR'S EQUATION

i) This model was unable to explain the spectrum of multi electronic species.

ii) No–scientific proof given for $\boxed{mv_nr_n = \frac{nh}{2\pi}}$

iii) This model could not explain "fine spectrum of H" means, when the line spectrum of H was analysed using high resolution spectroscope then it was observed that each line was actually combination of multiple closely space fine lines.

iv) In presence of electric field, each spectral line gets splitted. This phenomenon is called **Stark effect** and Bohr's model could not explain it.

v) In presence of magnetic field, spectral lines are splitted. This phenomenon is called **Zeeman's effect** and Bohr's model could not explain that.

iv) Later there were two theories which proved that the electron present in an atom have dominating wave nature while Bohr consider the electron as a particle only.

De–Broglie Hypothesis

It suggests the wave like nature of matter (particle).
With every moving object, a wave is always associated.
But a moving microscopic particle possess significant wave nature which can be observed experimentally.

(Davison – Germer experiment : Diffraction of electron beam)

(Significant nature means 10^{-11} m $\leq \lambda \leq 10^{-6}$ m)
These waves are called **Matter waves** and for a moving object, its wavelength can be calculated as following:

$$\boxed{\lambda = \frac{h}{p} = \frac{h}{mv}}$$

Where,
h = 6.6×10^{-34} Js
p = momentum of object (kg m/s)
m = mass of object (kg)
v = speed of object (m/s)
λ = de–Broglie wavelength (m)

Q49 **Calculate the wavelength associated**
(i) with an electron moving with a velocity of 10^6 m/sec.
(ii) with a cricket ball of mass 0.5 kg moving with 20 m/sec.

A49 Given velocity$_{e^-}$ = 10^6 m/sec

i) We know that,

$$\lambda = \frac{h}{mv}$$

$$\lambda = \frac{6.6 \times 10^{-34}}{(9.1 \times 10^{-31})10^6}$$

$$\lambda \simeq 7.27 \times 10^{-10}\ m$$

$$\lambda = 7.27\ Å$$

(Significant wave nature)

ii) Given mass = 0.5 kg and velocity = 20 m/sec

We know that,

$$\lambda = \frac{h}{mv}$$

$$\lambda = \frac{6.6 \times 10^{-34}}{0.5 \times 20}$$

$$\lambda = 6.6 \times 10^{-35} m$$ (i.e. Negligible wave nature)

Q50 **Calculate the momentum of a particle which has a de–Broglie wavelength of 3.31×10^{-10} m.**

A50 Given:

Wavelength,

$$\lambda = 3.31 \times 10^{-10} m$$

We know that :

$$\lambda = \frac{h}{p}$$

$$p = \frac{6.6 \times 10^{-34}}{3.31 \times 10^{-10}}$$

$$p = 2 \times 10^{-24}\ kgm/s$$

Q51 **Calculate the wavelength associated with an electron moving with kinetic energy 3.2×10^{-38} J.**

A51 Given kinetic energy = 3.2×10^{-38} J

For particle,

$$K.E = \frac{1}{2}mv^2$$

$$K.E = \frac{1}{2}mv^2 \times \frac{m}{m}$$

$$K.E = \frac{p^2}{2m}$$

$$p = \sqrt{2mK.E}$$

So,

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mK.E}}$$

K.E → Kinetic energy of object (Joule)

$$\lambda = \frac{6.6 \times 10^{-34}}{\sqrt{2 \times 9.1 \times 10^{-31} \times 3.2 \times 10^{-38}}}$$

$$\lambda = \frac{6.1 \times 10^{-34}}{\sqrt{582.4 \times 10^{-70}}}$$

$$\lambda = \frac{6.1 \times 10^{-34}}{24.13 \times 10^{-35}}$$

$$\lambda = 2.51 \text{ m}$$

Note: If a charged particle present at rest is accelerated by V volts, then **Increase in K.E = qV** where q is charge on particle

i.e. $\Delta KE = q.V$

$(KE)_f - (KE)_i = q.V$

So, $\boxed{\lambda = \frac{h}{\sqrt{2.m.q.V}}}$ Only for a charged particle initially at rest.

Q52 **A stream of electrons from a heated filament was passed between two charged plates kept at a potential difference V volt. If e and m are charge and mass of an electron respectively, then the value of $\frac{h}{\lambda}$ (where λ is wave-length associated with electron wave is given by:**

(A) $\sqrt{2meV}$ **(B)** meV **(C)** $2meV$ **(D)** $\sqrt{meV}$

A52 **A**

Given e and m are charge and mass of an electron

We know that:

$$\lambda = \frac{h}{\sqrt{2mqV}}$$

$$\frac{h}{\lambda} = \sqrt{2mqV}$$

$$\frac{h}{\lambda} = \sqrt{2meV}$$

Note: If an **electron** at rest is accelerated by V volts,

$$\lambda = \frac{h}{\sqrt{2mqV}}$$

$$\lambda = \frac{6.6 \times 10^{-34}}{\sqrt{2 \times 9.1 \times 10^{-31} \times 1.6 \times 10^{-19} V}}$$

$$\boxed{\lambda = \sqrt{\frac{150}{V}} \text{Å} = \frac{12.24}{\sqrt{V}}} \text{ Only for } e^-$$

Q53 **Calculate the de-Broglie wavelength of an electron accelerated through a potential difference of**
i) 75 volts **ii) 300 volts**

A53 i) Given potential difference = 75 volts

Mass of electron, m = 9.1×10^{31} kg

Charge of electron e = 1.6×19^{-19} C

$$\lambda = \frac{h}{\sqrt{2meV}}$$

$$\lambda = \frac{6.6\times10^{-34}}{\sqrt{2\times(9.1\times10^{-31})(1.6\times10^{-19})75}}$$

$$\lambda = \frac{6.6}{46.73}\times10^{-34+25}$$

$$\lambda = 0.141\times10^{-9}\ m$$

ii) Given potential difference V = 300 volts

We know that:

$$\lambda = \frac{h}{\sqrt{2meV}}$$

$$\lambda = \frac{6.6\times10^{-34}}{\sqrt{2\times(9.1\times10^{-31})(1.6\times10^{-19})\times300}}$$

$$\lambda = 0.07\times10^{-9}\ m$$

$\lambda_{matter} = \frac{h}{p}$

$\lambda_{matter} = \frac{h}{mv}$ } For all

$\lambda_{matter} = \frac{h}{2mK.E}$

$\lambda_{matter} = \frac{h}{2mqV}$ } Only for charged particles

$\lambda_{matter} = \sqrt{\frac{150}{V}}$ Å } Only for e^-

Note: In Bohr's model, for electron present in n^{th} orbit

$$v \simeq 10^6 m/s$$

$$m_e = 9.1 \times 10^{-31} kg$$

So, de – Broglie wavelength $\lambda = 10^{-10} m$

Also, Radius of orbit (r) $\approx 10^{-10}$ m

$\Rightarrow$ Circumference of orbit $\simeq 10^{-10}$ m

So, de–Broglie suggested that an electron present in a orbit is similar to a standing wave which keeps on oscillating w.r.t its orbit.

And the electron can be present in those orbit only for which the circumference is a natural number multiple of de–Broglie wavelength of the electron.

i.e. Circumference = n × de–Broglie wavelength

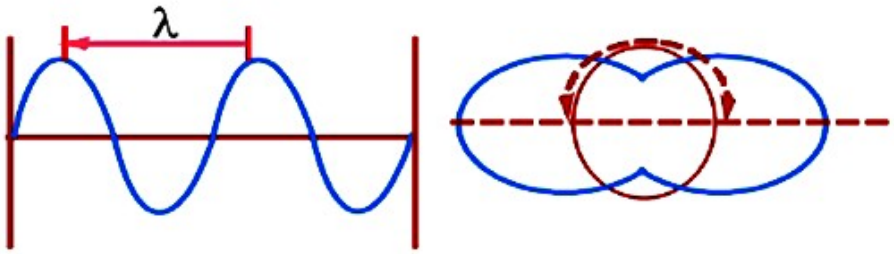
$$\boxed{2\pi r_n = n\lambda}$$

Where n = orbit number

For n=2 (second orbit)

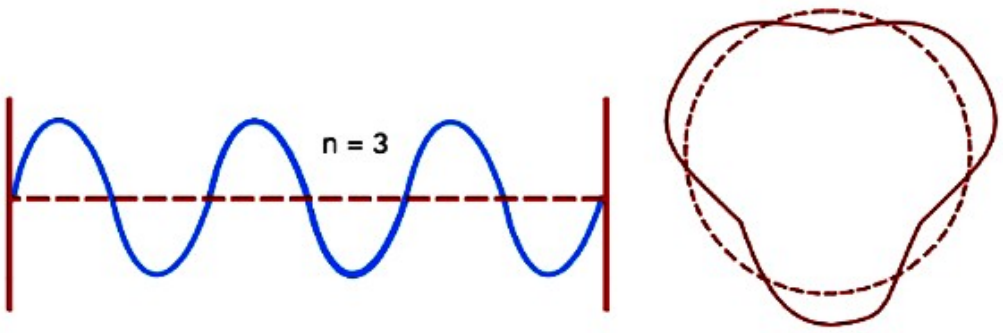
Circumference = $n\lambda$

Circumference = 2λ



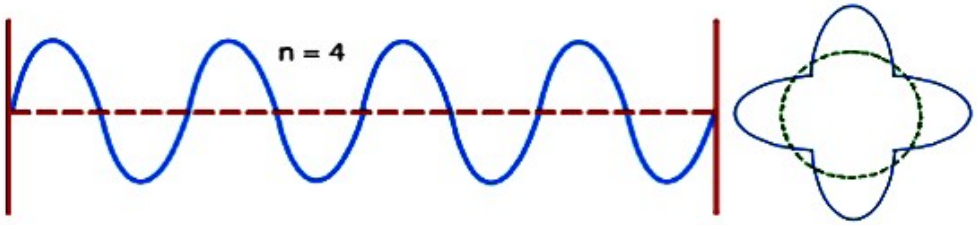
For n= 3

Circumference = $n\lambda = 3\lambda$



iii) For n = 4

Circumference = $n\lambda = 4\lambda$



Note : Proof for Bohr's equation

$$2\pi r_n = n\lambda$$

$$2\pi r_n = \frac{n \times h}{m v_n}$$

$$\Rightarrow \quad m v_n r_n = \frac{nh}{2\pi}$$

Heisenberg's Uncertainty Principle

LESSON-13

- HEISENBERG'S UNCERTAINTY PRINCIPLE
- UNCERTAINTY IN SPEED (Δv)
- UNCERTAINTY IN KINETIC ENERGY (ΔK)
- UNCERTAINTY IN DE-BROGLIE WAVELENGTH ($\Delta\lambda$),

(Uncertainty Means Error)

According to it, for a moving microscopic particle, it is impossible to determine accurate values of both position and momentum simultaneously.

It means if both of the these are determined simultaneously then their values have some error (uncertainty)

Let Δp is uncertainty (error) in momentum (kg m/s)

Δx is uncertainty in position (m)

So, in general

$$\boxed{\Delta p.\Delta x \geq \frac{h}{4\pi}} \quad \text{i.e.} \quad \begin{matrix}\Delta p \neq 0 \\ \Delta x \neq 0\end{matrix}$$

If there is uncertainty in momentum then all properties related to momentum will also have uncertainties (error)

i) Uncertainty in speed (Δv)

Since momentum = mass × velocity

or $p = mv$

Differentiate w.r.t v, $\frac{dp}{dv} = m$

$$dp = m.dv$$

$$\Delta p = m\Delta v$$

So, $$\Delta x \cdot \Delta p \geq \frac{h}{4\pi}$$

$$\Delta x(m\Delta v) \geq \frac{h}{4\pi}$$

$$\boxed{\Delta x.\Delta v \geq \frac{h}{4\pi m}}$$

Where, m is mass of particle.

ii) Uncertainty in Kinetic energy (ΔK)

Kinetic energy, $K = \frac{1}{2} mv^2$ ...(1)

Differentiate equation (1) w.r.t 'v' we get

$$\frac{dK}{dv} = \frac{1}{2} m(2v)$$

After rearrangement, we get

$\Rightarrow \quad \Delta K = mv.\Delta v$

So, $\Delta x.\Delta v \geq \dfrac{h}{4\pi m}$

$$\Delta x.\frac{\Delta K}{mv} \geq \frac{h}{4\pi m}$$

$$\boxed{\Delta x.\Delta K \geq \frac{hv}{4\pi}}$$

Where ΔK is uncertainty in KE(J) and v is speed of particle (m/s).

iii) Uncertainty in de–Broglie wavelength ($\Delta\lambda$)

From de–Broglie hypothesis we have,

$$\lambda = \frac{h}{p} \qquad \ldots(1)$$

Differentiating equation (1) w.r.t 'p' we get,

$$\frac{d\lambda}{dp} = \frac{-h}{p^2}$$

$$d\lambda = \frac{-h}{p^2}.dp$$

After differentiation, we get

$$|\Delta\lambda| = \frac{h}{p^2}.\Delta p$$

Using $\Delta x\ \Delta p \geq \dfrac{h}{4\pi}$

$$\Delta x\left(\frac{\Delta\lambda p^2}{h}\right) \geq \frac{h}{4\pi}$$

$$\Delta x.\Delta\lambda \geq \frac{\left(\frac{h}{p}\right)^2}{4\pi}$$

$$\boxed{\Delta x.\Delta\lambda \geq \frac{\lambda^2}{4\pi}}$$

iv) If the position of a particle is measured precisely. i.e. $\Delta x \to 0$ then $\Delta p \to \infty$.

If the momentum of the particle is measured precisely. i.e. $\Delta p \to 0$ then $\Delta x \to \infty$.

This is because of a principle of optics that if a light of wavelength 'λ' is used to locate the position of a particle then maximum error in the position measurement will be ±λ i.e. $\Delta x \to \pm\lambda$

If $\Delta x \to 0$; $\lambda \to 0$

But,

So, to make $\Delta x \to 0$, $\lambda \to 0$ a photon of very high energy is used to locate it.

∴ When this photon will collide with the electron then momentum of electron will get changed by a large amount.

$$\Delta p.\Delta x \geq \frac{h}{4\pi} \text{ (multiplied \& divided by } \Delta t)$$

$$\frac{\Delta p}{\Delta t}\Delta t.\Delta x \geq \frac{h}{4\pi} \qquad (\frac{\Delta p}{\Delta t} = \text{rate of change in momentum} = F)$$

$$F.\Delta x.\Delta t \geq \frac{h}{4\pi}$$

$$\Delta E.\Delta t \geq \frac{h}{4\pi}$$

$\Delta E \to$ uncertainty in energy

$\Delta t \to$ uncertainty in time

In terms of uncertainty in energy ΔE, and uncertainty in time Δt, this principle is written as,

$$\Delta E.\Delta t \geq \frac{h}{4\pi}.$$

Q54 **If the uncertainty in the position of an electron is 0.33 pm, what will be the uncertainty in its velocity?**

A54 Given, Uncertainty in position of an electron, $\Delta x = 0.33\ pm = 0.33\times10^{-12}\ m$

Using $\Delta x \Delta v \geq \frac{h}{4\pi m}$

$$\Delta v = \frac{6.626 \times 10^{-34}}{4 \times 3.14 \times 0.33 \times 10^{-12} \times 9.1 \times 10^{-31}} = 0.1754\times10^{-34+43}$$

$$= 0.1754\times10^{9} = 1.75\times10^{8}\ m/s$$

Q55 **If H^+ is accelerated to 6.62×10^6 m/s. If error in velocity is ± 1%, then find the uncertainty in position.**

A55 Given error in velocity = 1% of v

$$\Delta v = \frac{1}{100} \times 6.62 \times 10^{6} m/s$$

$$\Delta v = 6.62\times10^{4}\ m/s$$

Using, $\Delta x = \frac{h}{4\pi m \Delta v}$

$$\Delta x = \frac{6.6 \times 10^{-34}}{4 \times \frac{22}{7} \times 1.66 \times 10^{-27} \times 6.62 \times 10^{4}}$$

$$\Delta x = \frac{7 \times 6.6}{88 \times 1.66 \times 6.62} \times 10^{-11} m$$

$$\Delta x = 4.76 \times 10^{-13} m$$

Q56 **What is uncertainty in location of a proton of wavelength 5000 Å if wavelength is known to have an accuracy of 1 pm.**

(A) 7.96×10^{-14} m (B) 0.02 m (C) 3.9×10^{-8} m (D) None of these

A56 **B**

Given, wavelength, $\lambda = 5000Å$

$= 5000\times10^{-10}$ m

$= 5\times10^{-7}$ m

$\Delta\lambda = 1$ pm

$\Delta\lambda = 1\times10^{-12}$ m

$$\therefore \quad \Delta x.\Delta\lambda \geq \frac{\lambda^2}{4\pi}$$

$$\Delta x = \frac{\left(5\times10^{-7}\right)^2}{4\pi\times1\times10^{-12}}$$

$$\Delta x = \frac{25\times10^{-14}}{4\times3.14\times10^{-12}}$$

$$\Delta x = \frac{25\times10^{-14+12}}{12.56}$$

$$\Delta x = 1.99\times10^{-2}\ \text{m}$$

$$\Delta x = 0.0199\ \text{m}$$

$$\Delta x = 0.02\ \text{m}$$

Quantum – Mechanical Model / Wave Mechanical Model

LESSON–14

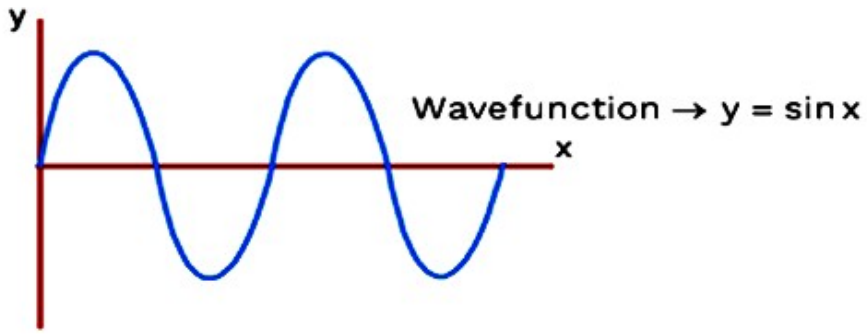
- QUANTUM – MECHANICAL MODEL / WAVE MECHANICAL MODEL
- SCHRODINGER EQUATION
- QUANTUM NUMBER

It is based upon quantum mechanics which was developed by Heisenberg and Schrodinger.
It is most correct model of atom till date but is mathematically complicated.

Important Features

It is impossible to detect the exact position of electron inside the atom. We can only determine the probability of finding electron at a certain point.

Electron doesn't exist in circular orbits or shells. But electron exists in orbitals which are 3–dimensional region of space in which probability of finding electron is maximum (≥95%).
The motion of electron around the nucleus is similar to a 3–D standing wave. So it should have a wavefunction which contains all information of the electron.



Schrodinger provided a tool to derive the wavefunction for a bound electron (within the atom) which is known as **Schrodinger equation**.

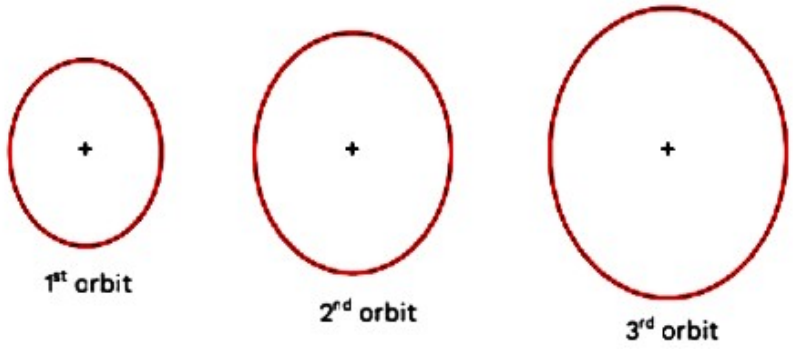
$$\frac{\partial^2\psi}{\partial x^2}+\frac{\partial^2\psi}{\partial y^2}+\frac{\partial^2\psi}{\partial z^2}+\frac{8\pi^2 m}{h^2}(E-V)\psi=0$$

Where x,y,z are cartesian coordinates
m is mass of electron
h is Planck's constant
E is Total energy of electron
V is potential energy of electron
ψ is wave function of electron
The above equation is a "second order differential equation" which when solved for a bound electron, gives wave function.

When above equation is solved for a bound electron, three unique constants are obtained which give values of three quantum numbers out of four i.e.

(i) Principal quantum number
(ii) Azimuthal quantum number } From solution of Schrodinger equation
(iii) Magnetic quantum number
(iv) Spin quantum number } Theortically defined

An orbit or shell represents the locus of all those point present at a certain distance from the nucleus, beyond which electron can not go.



i.e. orbits are co-centric spheres.
In an atom, orbits are distinguished by Principal quantum number (n).
i.e. If n=1 ⇒ 1st orbit
n=2 ⇒ 2nd orbit
n=3 ⇒ 3rd orbit
Every shell or orbit consists one or more type of subshells which can be distinguished by Azimuthal quantum number
i.e. If l=0 ⇒ s-subshell
l=1 ⇒ p-subshell
l=2 ⇒ d-subshell
l=3 ⇒ f-subshell

	Representation
1st Orbit → Only s	1s
2nd Orbit → s,p	2s,2p

3rd Orbit → s,p,d	3s,3p,3d
4th Orbit → s,p,d,f	4s,4p,4d,4f

Each subshell contains a certain number of orbitals which are distinguished by Magnetic quantum number (m).

For a particular subshell
No. of orbitals = (2l + 1)

Subshell	No. of orbitals	Values of m
s(l=0)	1	{0}
p(l=1)	3	{−1,0,1}
d(l=2)	5	{−2,−1,0,1,2}
f(l=3)	7	{−3,−2,−1,0,1,2,3}

An orbital can accommodate maximum of 2 electrons which are distinguished by spin quantum number(s).

$$s = \left\{+\frac{1}{2} \text{ or } -\frac{1}{2}\right\}$$

Note:

Each set of <n, l, m, s> values represents one electron uniquely.

Quantum Numbers:

The set of four numbers required to define an electron completely in an atom are called quantum numbers. The first three have been derived from Schrodinger wave equation.

(i) Principal quantum number (n) : (Proposed by Bohr)

It describes the size of the electron wave and the total energy of the electron. It has integral values 1, 2, 3, 4 ...., etc., and is denoted by K, L, M, N. ..., etc.

- Number of subshell present in n^{th} shell = n

N	subshell
1	s
2	s, p
3	s, p, d
4	s, p, d, f

- Number of orbitals present in nth shell = n^2.
- The maximum number of electrons which can be present in a principal energy shell is equal to $2n^2$. No energy shell in the atoms of known elements possesses more than 32 electrons.
- Angular momentum of electron of any orbit = $\frac{nh}{2\pi}$

(ii) Azimuthal quantum number (l) : (Proposed by Sommerfield)

It describes the shape of electron cloud and the number of subshells in a shell.

- It can have values from 0 to (n – 1).

Value of l	Subshell
0	s
1	p
2	d
3	f

- Number of orbitals in a subshell = 2l + 1
- Maximum number of electrons in particular subshell = 2 × (2l + 1)
- Orbital angular momentum L = $\frac{h}{2\pi}\sqrt{l(l+1)} = \hbar\sqrt{l(l+1)}$ $\left[\hbar = \frac{h}{2\pi}\right]$

i.e. Orbital angular momentum of s orbital = 0

Orbital angular momentum of p orbital = $\sqrt{2}\frac{h}{2\pi}$

Orbital angular momentum of d orbital = $\sqrt{3}\frac{h}{2\pi}$

(iii) Magnetic quantum number (m) : (Proposed by Linde)

It describes the orientations of the subshells. It can have values from –l to +l including zero, i.e., total (2l + 1) values. Each value corresponds to an orbital. s-subshell has one orbital, p-subshell has three orbitals (px, py and pz), d-subshell has five orbitals (d_{xy}, d_{yz}, d_{zx}, $d_{x^2-y^2}$, d_{z^2}) and f-subshell has seven orbitals. The total number of orbitals present in a main energy level is 'n2'.

(iv) Spin quantum number (s) : (Proposed by Samuel Goldsmit & Uhlenbeck)

It describes the spin of the electron. It has values +1/2 and –1/2. signifies clockwise spinning and anticlockwise spinning.

- Spin magnetic moment $= \frac{eh}{2\pi mc}\sqrt{s(s+1)}$ or $\mu = \sqrt{n(n+2)}$ B.M. (n = number of unpaired electrons)
- It represents the value of spin angular momentum which is equal to $\frac{h}{2\pi}\sqrt{s(s+1)}$.
- Maximum spin of atom = $\frac{1}{2}\times$ No. of unpaired electron.

Electronic configuration:

Pauli's exclusion principle:

No two electrons in an atom can have the same set of all the four quantum numbers, i.e., an orbital cannot have more than 2 electrons because three quantum numbers (principal, azimuthal and magnetic) at the most may be same but the fourth must be different, i.e., spins must be in opposite directions.

Aufbau principle:

Aufbau is a German word meaning building up. The electrons are filled in various orbitals in order of their increasing energies. An orbital of lowest energy is filled first. The sequence of orbitals in order of their increasing energy is:

1s, 2s, 2p, 3s, 3p, 4s, 3d, 4p, 5s, 4d, 5p, 6s, 4f, 5d, 6p, 7s, 5f, 6d, ...

The energy of the orbitals is governed by (n + l) rule.

'n + l' Rule:

The relative order of energies of various sub-shell in a multi electron atom can be predicated with the help of 'n + l' rule.

- The sub-shell with lower value of (n + l) has lower energy and it should be filled first.

e.g. 3d 4s

$(n+l) = 3+2 = 5$ $(n+l) = 4+0 = 4$

Since, (n + l) value of 3d is more than 4s therefore, 4s will be filled before 3d.

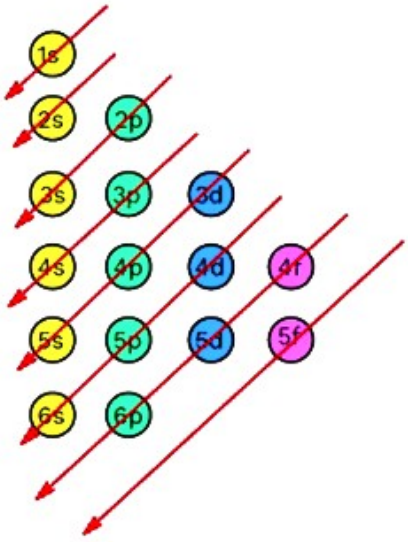
- If two sub-shell has same value of (n + l) then the sub-shell with lower value of n has lower energy and it should be filled first.

e.g. 3d 4p

$(n+l) = 3+2 = 5$ $(n+l) = 4+1 = 5$

Since, (n + l) value of 3d and 4p is same, 3d will be filled before 4p.

Memory Map:



1 - s
2 - s, p
3 - s, p
4 - s, p, d
5 - s, p, d
6 - s, p, d, f
7 - s, p, d, f

Hund's Rule:

No electron pairing takes place in the orbitals in a sub - shell until each orbital is occupied by one electron with parallel spin. Exactly half filled and fully filled orbitals make the atoms more stable, i.e., p^3, p^6, d^5, d^{10}, f^7 and f^{14} configuration are most stable.

Q57 **Write the electronic configuration and find the no. of unpaired electrons as well as total spin for the following atoms:**

(i) $_{6}C$
(ii) $_{8}O$
(iii) $_{15}P$
(iv) $_{21}Sc$
(v) $_{26}Fe$
(vi) $_{10}Ne$

A57 (i) $_{6}C \rightarrow 1s^2\ 2s^2\ 2p^2$



Number of unpaired electrons → 2

Total spin = $\frac{+2}{2}$ or $\frac{-2}{2}$

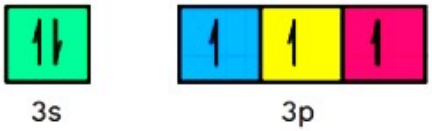
(ii) $_{8}O \rightarrow 1s^2\ 2s^2\ 2p^4$



Number of unpaired electrons → 2

Total spin = $\frac{+2}{2}$ or $\frac{-2}{2}$

(iii) $_{15}P \rightarrow 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^3$

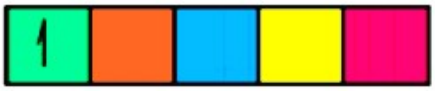


Number of unpaired electrons → 3

Total spin = $\frac{+3}{2}$ or $\frac{-3}{2}$

(iv) $_{21}Sc \rightarrow 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^1$

or $[Ar]4s^2 3d^1$



3d

$[Ar]^3d^1 4s^2$

$\therefore$ Number of unpaired electrons $\rightarrow 1$

$\therefore$ Total spin = $\frac{+1}{2}$ or $\frac{-1}{2}$

(v) $_{26}Fe \rightarrow 1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^6$

$[Ar]4s^2\ 3d^6$



4s 3d

$\therefore$ Number of unpaired electrons = 4

$\therefore$ Total spin = $\frac{+4}{2}$ or $\frac{-4}{2}$

(vi) $_{10}Ne \rightarrow 1s^2\ 2s^2\ 2p^6$



Number of unpaired electrons = 0

Total spin = 0

Q58 **Write down the four quantum numbers for fifth and sixth electrons of carbon atom.**

A58 6C : $1s^2\ 2s^2\ 2p^2$

Fifth electron : n = 2, l = 1, m = –1 or +1 , s = +1/2 or –1/2

Sixth electron : n = 2, l = 1, m = 0, s = +1/2 or –1/2

Q59 **Calculate total spin, magnetic moment for the atoms having atomic number 7, 24 and 36.**

A59 The electronic configuration are

$_{7}N$: $1s^2\ 2s^2\ 2p^3$ Unpaired electron = 3

$_{24}Cr$: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^5\ 4s^1$ Unpaired electron = 6

$_{36}Kr$: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^2\ 4p^6$ Unpaired electron = 0

$\therefore$ Total spin for an atom = ± 1/2 × no. of unpaired electron

For $_{7}N$, it is = ± 3/2 ; For $_{24}Cr$, it is = ± 3 ; For $_{36}Kr$, it is = 0

Also magnetic moment = $\sqrt{n(n+2)}$

For $_{7}N$, it is = $\sqrt{15}$; For $_{24}Cr$, it is $\sqrt{48}$; For $_{36}Kr$, it is = $\sqrt{0}$

Exceptions:

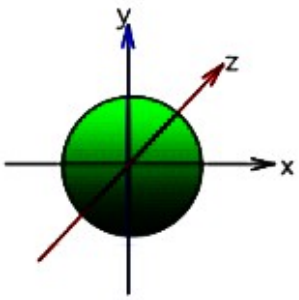
(1) $_{24}Cr$ = [Ar] $4s^2\ 3d^4$ (Not correct)
[Ar] $4s^1\ 3d^5$ (correct: as d^5 structure is more stable than d^4 structure)

(2) $_{29}Cu$ = [Ar] $4s^1\ 3d^{10}$ (correct: as d^{10} structure is more stable than d^9 structure).

Shape of the orbitals:

Shape of the orbitals are related to the solutions of Schrodinger wave equation, and gives the space in which the probability of finding an electron is maximum.

s-orbital: Shape → spherical

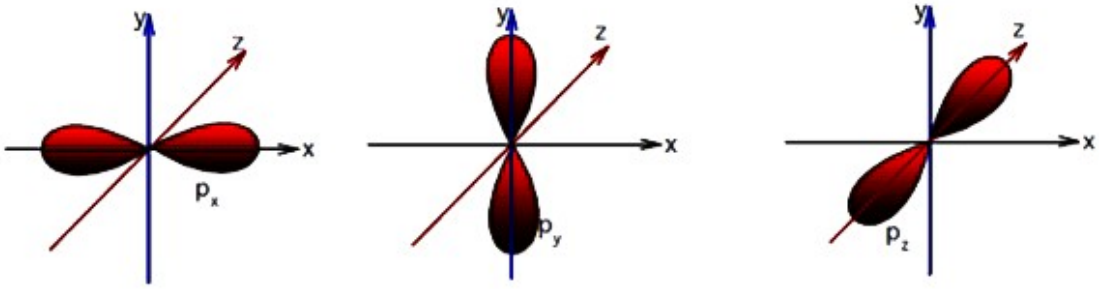


s-orbital is non directional and it is closest to the nucleus, having lowest energy.

s-orbital can accommodate maximum number of two electrons.



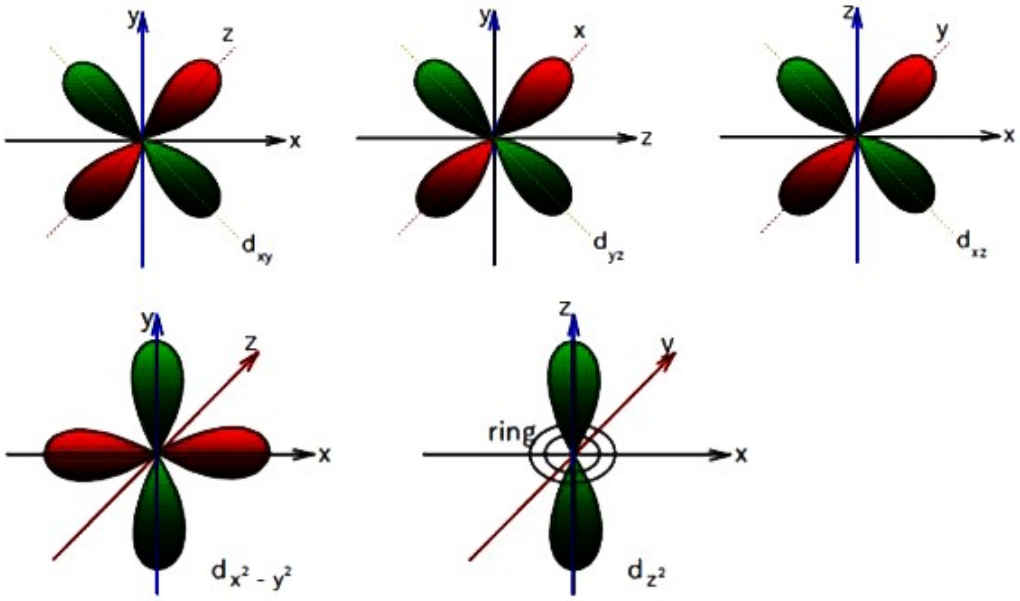
p-orbital: Shape → dumb bell



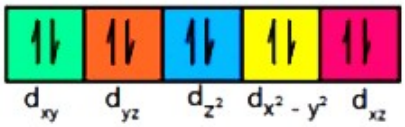
p-orbital can accommodate maximum number of six electrons.



d-orbital: Shape → double dumb bell



d-orbital can accommodate maximum number of ten electrons.



f-orbital: Shape → leaf like

f-orbital can accommodate maximum number of fourteen electrons.



Introduction to Nodes

LESSON–15

- INTRODUCTION TO NODES
- SCHRODINGER'S EQUATION FOR H–ATOM
- INTERPRETATION OF ψ
- RADIAL NODE (RN)
- ANGULAR NODES (AN)

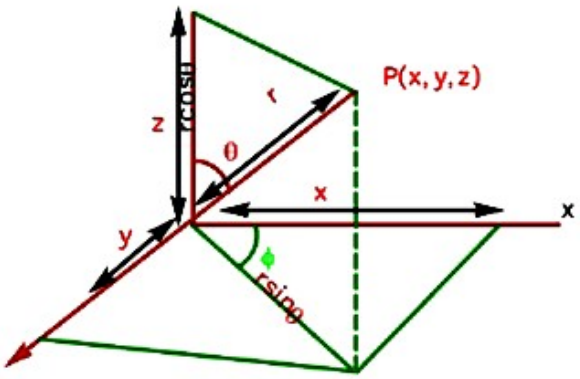
Schrodinger's Equation For H–Atom

When Schrodinger's equation is solved for a bound electron, the wave function is obtained as a functions of x, y & z.

i.e. $\psi = F(x, y, z)$

But Mathematically it is very complicated to solve in certain co–ordinates. So convert them into polar co–ordinates (r, θ, ϕ)

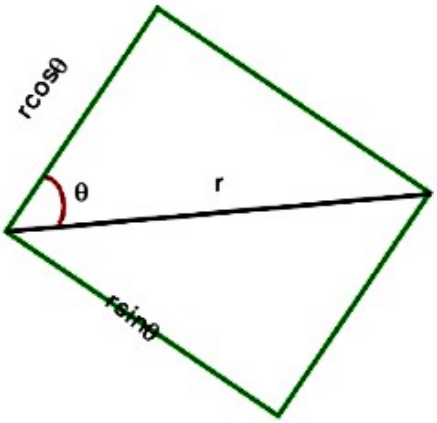
Consider a point P in 3–D space as following:

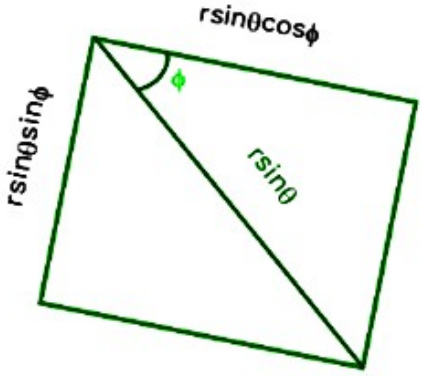


i.e. $x = r\sin\theta\cos\phi$

$y = r\sin\theta\sin\phi$

$z = r\cos\theta$





So, when Schrodinger equation is solved in polar co–ordinates, ψ of electron is obtained as following:

$$\psi = F(r, \theta, \phi)$$

More correctly,

$$\boxed{\psi = R(r).A(\theta, \phi)}$$

i.e. as the multiplication of two independent equation

$R(r) \rightarrow$ Radial part of wavefunction
$A(\theta, \phi) \rightarrow$ Angular part of wavefunction

Interpretation of ψ

ψ is known as probability amplitude of the electron
$R(r) \rightarrow$ Radial probability amplitude
$A(\theta,\phi) \rightarrow$ Angular probability amplitude
ψ can be positive, negative or zero.
There is no physical significance of ψ
But ψ^2 has a significance.
$\psi^2 \rightarrow$ Probability density
It is defined as the probability of finding electron per unit volume.

i.e. $\boxed{\psi^2 = \frac{\text{Probability}}{\text{Volume}}}$

So, for a given volume (dv)

$$\boxed{\text{Probability} = \psi^2 dv}$$

Since ψ has two components so ψ^2 will also have two components as following

i) **$R^2(r) \rightarrow$ Radial probability density**
It is probability per unit volume at a certain distance from the nucleus in all directions.
i.e. probability density on a spherical surface.

ii) **$A^2(\theta,\phi) \rightarrow$ Angular probability density**
It is probability per unit volume in a particular direction at all possible radial distance.

Conclusions

Every s–orbital is spherical and non–directional so their wavefunctions do not contains angular part.

i.e. $\boxed{\psi_{s\text{-orbital}} = R(r) \text{ only}}$

Since all p, d and f orbitals are directional so their wavefunction contains angular part.
The locus of all point where probability of finding electron is zero, is called as **Node**

i.e. at node: Probability = 0

$\Rightarrow \psi^2 dv = 0$

$\Rightarrow \psi^2 = 0$

$\Rightarrow \psi = 0$

$\Rightarrow R(r).A(\theta,\phi)=0$

So there are two types of nodes as following:

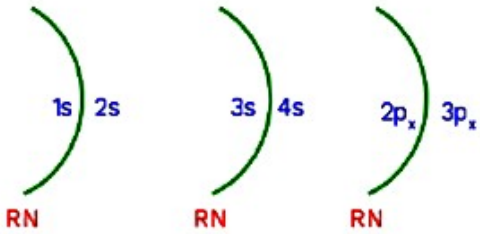
1. Radial node (RN)

The locus of all points where radial part of wavefunction is zero.

i.e. **R(r) = 0**

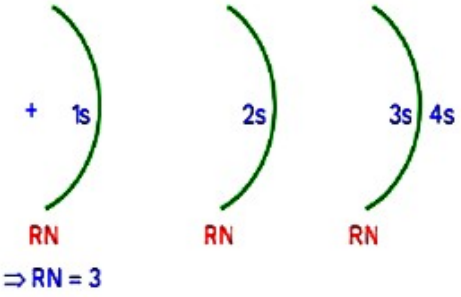
These are like spherical surfaces.

Same type orbitals of two successive shells are separated by one radial node.

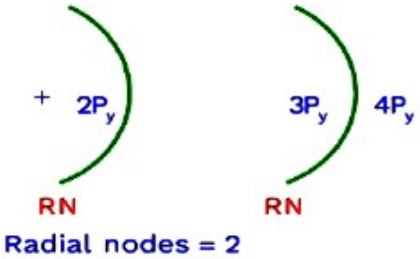


For a given orbital, total number of radial nodes are counted from nucleus.

e.g. i) 4s–orbital



4p$_y$ orbital



2. Angular Nodes (AN)

Angular nodes means nodal plane.

Locus of all points where $A(\theta,\phi) = 0$

These are 2–D planes as following.

Orbital	AN
s–orbital	N/A
p_x	YZ – plane
p_y	XZ – plane
pz	XY – plane

Note: For a given orbital

No. of Radial node $= (n - l - 1)$

No. of Angular node $= (l)$

Total nodes $=$ Radial node + Angular node $= n - 1$

e.g. 4s

Radial Node = (4–0–1) = 3

Identification of orbital from wavefunction

LESSON–16

- IDENTIFICATION OF ORBITAL FROM WAVEFUNCTION
- PROBLEMS ON WAVEFUNCTIONS
- RULES FOR "R(r) VS r" GRAPH

Each wavefunction contains an exponential term as following:

$$e^{\frac{-Zr}{n.a_0}}$$

Where,

Z is atomic number

r is radial distance from nucleus

n is principal quantum number

a_0 is Bohr radius of 1st orbit of H

For any electron:

$$R(r) \propto \left(\frac{1}{a_0}\right)^{\frac{3}{2}+\ell}$$

Where ℓ is Azimuthal quantum number.

For any electron:

If R(r) = 0

Number of finite solution = No. of Radial node

= (n – l – 1) (except r = 0 and r = ∞)

Q60 **Identify orbitals for H–atom corresponding to following wavefunction:**

A60

(i) $\psi = \frac{1}{\sqrt{\pi} a_0^{\frac{3}{2}}} e^{\frac{-r}{a_0}}$

Given, $e^{\frac{-r}{a_0}} = e^{\frac{-Zr}{na_0}}$

$e^{\frac{-r}{a_0}} = e^{\frac{-1r}{na_0}}$

On comparing we get

$\Rightarrow \quad \frac{-r}{a_0} = \frac{-1r_0}{na_0}$

$\boxed{n = 1}$

We know that:

$R(r) \propto \left(\frac{1}{a_0}\right)^{\frac{3}{2}+\ell}$

$R(r) \propto \left(\frac{1}{a_0}\right)^{\frac{3}{2}} \qquad [\Rightarrow \ell = 0], \qquad$ s – orbital

i.e. 1s orbital

ii) $\psi = \frac{1}{4\sqrt{2\pi} a_0^{\frac{3}{2}}} \left(2 - \frac{r}{a_0}\right) e^{-\frac{r}{2a_0}}$

Given, No terms of θ and ϕ means s orbital

$e^{\frac{-r}{2a_0}} = e^{-\frac{r}{na_0}}$

$\Rightarrow \quad \frac{-r}{2a_0} = \frac{-r}{na_0}$

$\boxed{n = 2}$

$\psi \propto \left(\frac{1}{a_0}\right)^{\frac{3}{2}}$ only

$\Rightarrow \quad \ell = 0$

$\boxed{\text{2s orbital}}$

i.e. $n - \ell - 1 = 1$

$$R(r) = 0$$

$$\left(2 - \frac{r}{a_0}\right) = 0$$

$$r = 2a_0$$

i.e. only one RN at a distance of $2a_0$ from nucleus

iii) $$\psi = \frac{\sqrt{2}}{81\sqrt{\pi}a_0^{3/2}}\left(6 - \frac{r}{a_0}\right)\left(\frac{r}{a_0}\right)e^{-\frac{r}{3a_0}}\cos\theta$$

$$e^{-\frac{r}{3a_0}} = e^{-\frac{nr}{na_0}}$$

$$-\frac{r}{3a_0} = \frac{-r}{na_0}$$

$$\Rightarrow n = 3$$

3p orbital

$$R(r) \propto \left(\frac{1}{a_0}\right)^{\frac{3}{2}+\ell}$$

$$\Rightarrow \ell = 1$$

$R(r) = 0$

$\Rightarrow$ $r = 6a_0$

i.e. RN = 1 = $(n - \ell - 1)$

iv) $$R(r) = \frac{1}{81\sqrt{\pi}a_0^{3/2}}\frac{r^2}{a_0^2}e^{\frac{-r}{3a_0}}$$

$$e^{\frac{-r}{3a_0}} = e^{\frac{-nr}{na_0}}$$

$$\Rightarrow n = 3$$

3d orbital

Also

$$R(r) \propto \left(\frac{1}{a_0}\right)^{\frac{3}{2}+2}$$

$$\Rightarrow \ell = 2$$

v) $$R(r) = \frac{1}{81\sqrt{3\pi}a_0^{3/2}}\left(27 - 18\sigma + 2\sigma^2\right)e^{\frac{-\sigma}{3}}$$

Where $\sigma = \frac{r}{a_0}$

$$e^{\frac{-r}{3a_0}} = e^{\frac{-nr}{na_0}}$$

$$\frac{-r}{3a_0} = \frac{-r}{na_0}$$

$$\Rightarrow \boxed{n = 3}$$

3s orbital

$$R(r) \propto \left(\frac{1}{a_0}\right)^{\frac{3}{2}}$$

$$\Rightarrow \ell = 0$$

$R(r) = 0 \Rightarrow$ 2 finite solution

R.N = 2 = (n–l–1)

Q61 **Determination of radial nodes from wavefunction.**

(i) $$R_{(r)} = \frac{1}{81\sqrt{3}\pi a_0^{3/2}}\left(27 - 18\sigma + 2\sigma^2\right)e^{\frac{-\sigma}{3}}$$

Where $\sigma = \frac{r}{a_0}$

A61 Given, R(r)=0

$$2\sigma^2 - 18\sigma + 27 = 0$$

$$\sigma = \frac{-(-18) \pm \sqrt{324 - (216)}}{4}$$

$$\sigma = \frac{18 \pm 2\sqrt{27}}{4}$$

$$\frac{r}{a_0} = \frac{9 \pm \sqrt{27}}{2} \qquad \left[\therefore \sigma = \frac{r}{a_0} \text{ Given}\right]$$

i.e. $$r = \left(\frac{9 + 3\sqrt{3}}{2}\right)a_0$$

$$r = \left(\frac{9 - 3\sqrt{3}}{2}\right)a_0$$

Two radial nodes

Rules for "R(r) vs r" graph

Nucleus is present at the origin.

For any s–orbital, the curve starts from a maximum value.

For any p, d, f orbital, the curve starts from origin.

The point where the curve intersects with x–axis, represents a radial node.

i.e.

No. of radial nodes = Number of cuts on x – axis

After each node, the sign of ψ changes.

As 'r' increases, height of successive peaks decrease.

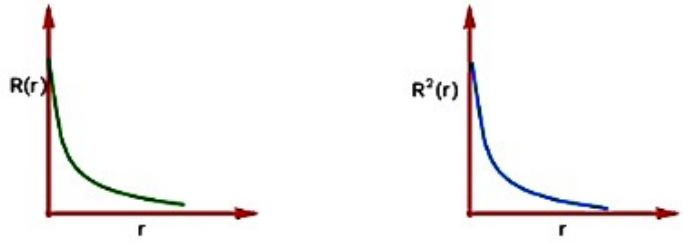
At the end, the curve will be parallel to x–axis.

For "$R^2(r)$ vs r" graph, the curve is always positive with same positions of RN.

"R(r) vs r" graph

Example:

i) For 1s orbital

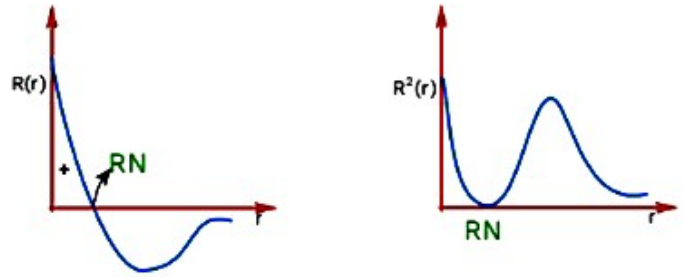


Starting from maximum

⇒ RN = (1–0–1) = 0

∴ Intersection = 0

ii) For 2s orbital

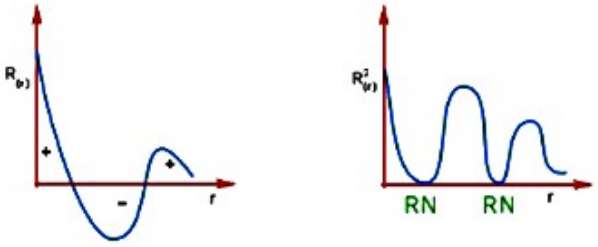


Starting from maximum

⇒ RN = (2–0–1) = 1

∴ Cuts = 1

iii) For 3s orbital



Starting from maximum

⇒ RN = (n–l–1)

⇒ RN = 3–0–1

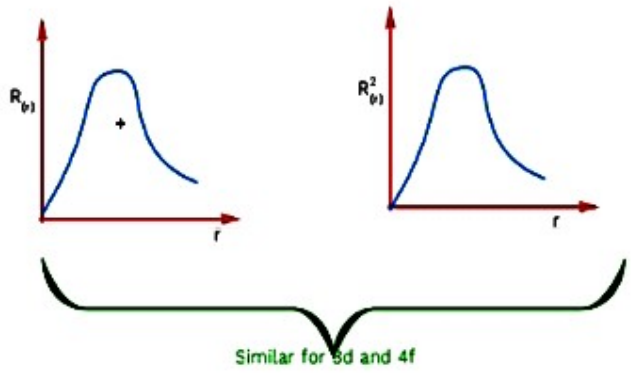
⇒ RN = 2

∴ Cuts = 2

CONTENT

- R(r) Vs r Graph
- RADIAL PROBABILITY DISTRIBUTION FUNCTION (RPDF)
- RULES FOR $P_{(r)}$ Vs r GRAPH
- PROBLEMS ON RPDF

iv) For 2p orbital



Starting from origin

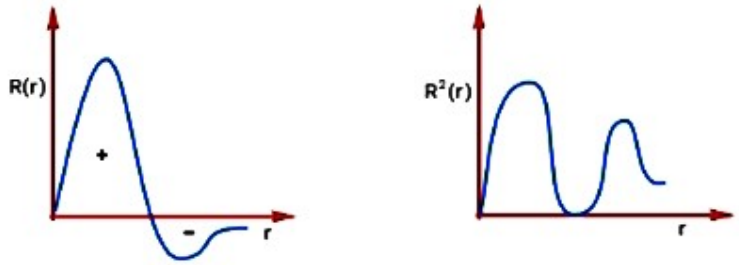
$\Rightarrow$ RN = (n – l – 1)

$\Rightarrow$ RN = (2 – 1 – 1)

$\Rightarrow$ RN = 0

$\therefore$ Cuts = 0

v) For 3p (or 4d or 5f)

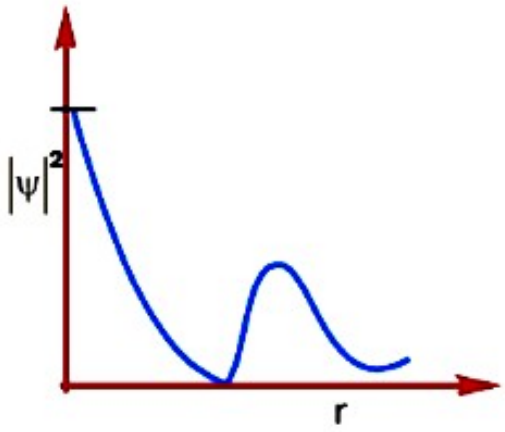


Starting from origin

$\Rightarrow$ R.N = (3–1–1) = 1

$\therefore$ Cuts = 1

Q62 **The graph between $|\psi|^2$ and r(radical distance) is shown below. This represents:**



(A) 2s orbital **(B) 2p orbital** **(C) 3s orbital** **(D) 1s orbital**

A62 **(A)**

It is clear from the graph it has one node and starting from maximum i.e. 2s orbital.

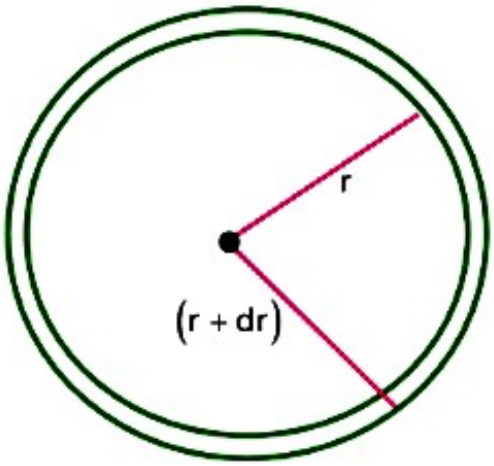
Radial Probability Distribution Function (RPDF)

It is a function which gives the distribution of radial probability, means the probability of finding electron at a particular distance in all directions. Means this gives the probability of finding electron over a spherical surface of radius(r).

Probability on surface = Probability density × $(Volume)_{spherical\ surface}$

Lets consider a spherical surface of radius 'r' having thickness dr (dr <<<< r)

$(Volume)_{Surface}$ = area × thickness = $(4\pi r^2)dr$



i.e. Radial probability = $(R^2(r)\ 4\pi r^2)dr$

$$\boxed{\left(R^2(r).4\pi r^2\right)dr = RPDF} = P(r)$$

Probability of electron at a distance 'r'

Rules for "$P_{(r)}$ vs r" graph

Nucleus is present at the origin.

For any orbital, the curve starts from origin.

$\because P_{(r)} \alpha\ r^2$

So at r =0 $\Rightarrow$ P(r) = 0

For any orbital, the curve is present in 1st quadrant only.

$\therefore \quad P_{(r)} = R^2_{(r)}.\ 4\pi r^2 > 0$

Each touch with x axis, represents a radial node.

i.e. Number of Radial node = Number of touches with x–axis (origin is not considered or counted in touches).

Number of peaks = (no of touches +1).

As small 'r' increases, height of successive peaks increases.

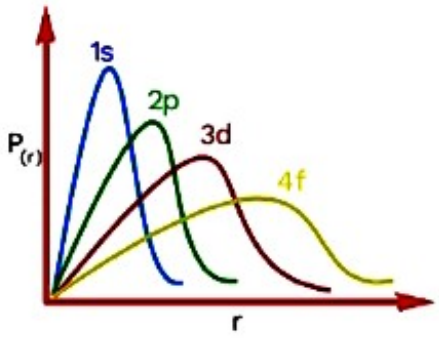
Example:

i) For 1s (or 2p or 3d or 4f)

Starting from origin

Also RN = 0

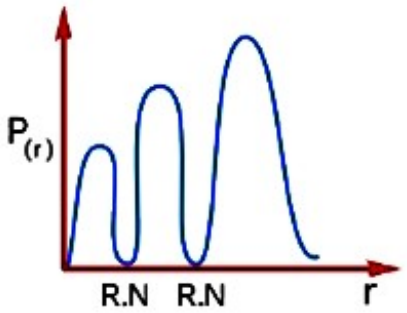
No. of peaks = 1



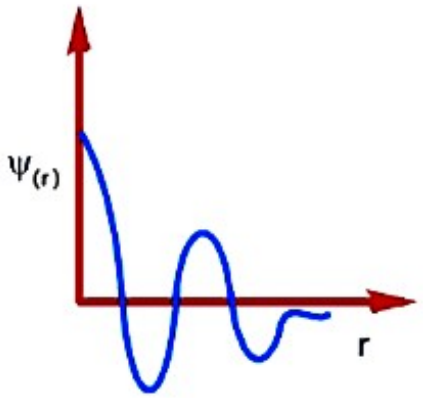
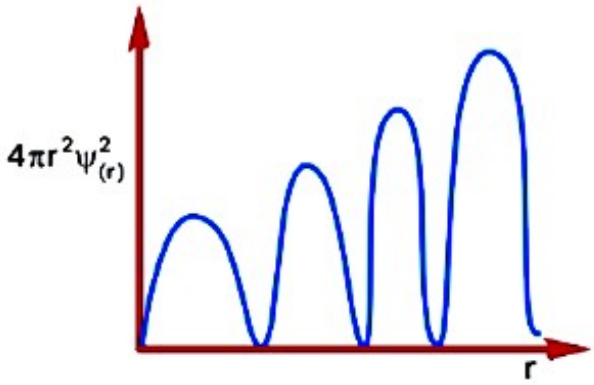
ii) For 3s/4p/5d/6f

Radial nodes = 2

No of peaks = 3



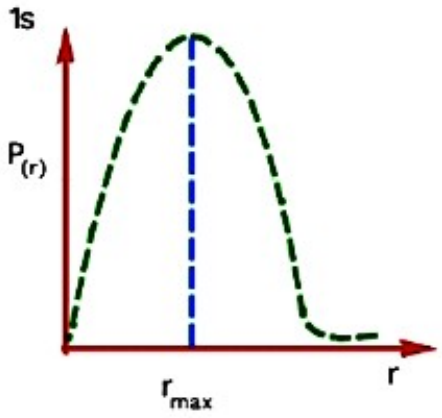
Q63 **Column I and Column II contain data on Schrodinger Wave–Mechanical Model, where symbol have their usual meanings match the columns**

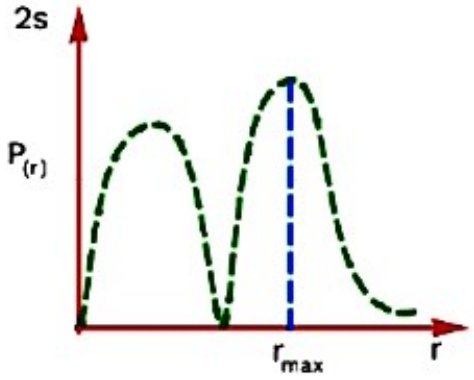
Column – I		Column – II (Type of orbital)	
(A)		(P)	4s
(B)		(Q)	Any of the 5p orbital
(C)	$\psi(\theta,\phi)$ = K (independent of θ and ϕ)	(R)	3s
(D)	At least one angular node is present	(S)	Any of the 6d orbital

A63 A→(P), B→ (P,Q,S), C→(P,R), D→(Q,S)

r_{max} (The Distance Corresponding to the Highest peak)

The radial distance from nucleus where probability of finding electron is maximum.





For calculation

$$\boxed{\frac{dP(r)}{dr} = 0}$$

Largest r= r_{max}

e.g. for 1s of H–atom

$$P_{(r)} = R^2(r).4\pi r^2 = \left(\frac{1}{\sqrt{\pi}a_0^{3/2}} e^{-\frac{r}{a_0}}\right)^2 .4\pi r^2$$

$$P_{(r)} = \frac{4}{a_0^3} e^{\frac{-2r}{a_0}} r^2$$

$$\frac{dP_{(r)}}{dr} = \frac{4}{a_0^3}\left[e^{\frac{-2r}{a_0}} 2r + r^2\left(\frac{-2}{a_0}\right)e^{\frac{-2r}{a_0}}\right] = 0$$

$\Rightarrow$ $r = a_0 \Rightarrow r_{max} = 0.529$ Å

Note : The Bohr radius is actually r_{max}

